STAT 216 Coursepack

Fall 2026
Montana State University

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This resource was developed by Melinda Yager, Jade Schmidt, and Stacey Hancock in 2021 to accompany the online textbook: Hancock, S., Carnegie, N., Meyer, E., Schmidt, J., Yager, M., and Greenwood, M. (2026). *Montana State Introductory Statistics with R*. Montana State University. https://mtstateintrostats.github.io/IntroStatTextbook/.

Contents

Preface

This coursepack accompanies the textbook for STAT 216: Montana State Introductory Statistics with R, which can be found at https://mtstateintrostats.github.io/IntroStatTextbook/. The syllabus for the course (including the course calendar), data sets, and links to Canvas, Gradescope, and the MSU RStudio server can be found on the course webpage: https://math.montana.edu/courses/s216/. Other notes and review materials are linked in Canvas.

Each of the activities in this workbook is designed to target specific learning outcomes of the course, giving you practice with important statistical concepts in a group setting with instructor guidance. In addition to the in-class activities for the course, video notes are provided to aid in taking notes while you complete the required videos. Bring this workbook with you to class each class period, and take notes in the workbook as you would your own notes. A well-written completed workbook will provide an optimal study guide for exams!

All activities and labs in this coursepack will be completed during class time. Parts of each lab will be turned in on Gradescope. To aid in your understanding, read through the introduction for each activity before attending class each day.

Additional review activities for each Module are provided at the end of each Unit. These review activities will be completed during weekly review sessions throughout the semester.

STAT 216 is a 3-credit in-person course. In our experience, it takes six to nine hours per week outside of class to achieve a good grade in this class. By "good" we mean at least a C because a grade of D or below does not count toward fulfilling degree requirements. Many of you set your goals higher than just getting a C, and we fully support that. You need roughly nine hours per week to review past activities, read feedback on previous assignments, complete current assignments, and prepare for the next day's class. A typical week in the life of a STAT 216 student looks like:

- *Prior to class meeting*:
  - Read assigned sections of the textbook, using the provided reading guides to take notes on the material.
  - Watch the provided videos, taking notes in the coursepack.
  - Read through the introduction to the day's in-class activity.
  - Read through the week's homework assignment and note any questions you may have on the content.
- *During class meeting*:
  - Work through the activities and labs with your classmates and instructor, taking detailed notes on your answers to each question in the activity.
- *After class meeting*:
  - Complete any parts of the activity you did not complete in class.
  - Review the activity solutions in the Math and Stat Center, and take notes on key points.
  - Complete any remaining assigned readings videos, and quizzes for the week.
  - Complete the week's homework assignment.
  - Attend weekly review sessions or complete the provided review worksheets

MODULE 1

Basics of Data and Sampling Methods

1.1 Vocabulary Review and Key Topics

At the beginning of each module is a summary of key topics and new vocabulary terms for that module. As you read through the material in the textbook and watch the videos prior to class, look for these terms. Reference the definitions to guide your understanding.

1.1.1 Key topics

Module 1 introduces the foundations of data: observational units, types of variables, and how to collect sample data from a population of interest in a way that allows us to generalize our results back to the population.

1.1.2 Vocabulary

- **Data**: observations used to answer research questions.
- **Observational units (cases)**: the subjects or entities on which data are collected.
  - The rows in a data set represent the observational units.
- **Sample size**: the number of observational units in a data set, denoted by n.
- **Variable**: the characteristics collected on each observational unit.
- **Types of variables**:
  - **Categorical**: cases are grouped into categories.
  - **Quantitative**: numerical measurements, where performing arithmetic operations makes sense.
- **Target population**: group of observational units of interest.
- **Sample**: subset of the population.
- **Statistic**: numerical value calculated on a sample; for categorical data, a proportion is calculated, for quantitative data, the mean is calculated.
- **Parameter**: numerical value of the entire population we are interested in; this is generally an unknown value.
- **Sampling methods**:
  - **Unbiased sampling method (e.g., a random sample)**: on average, the sample will be representative of the target population; all observational units in the target population have the same chance of being selected.
  - **Biased sampling method (e.g., convenience sample)**: on average, the sample will not be representative of the target population; some part of the target population will be over- or under-represented.
- **Type of sampling bias**:
  - **Selection bias**: method of sampling is biased; some part of the target population is over- or under-represented.

- **Non-response bias**: part of a pre-selected sample does not respond or cannot be reached.
- **Response bias**: responses are not truthful (poor/leading question phrasing, social desirability).

- **Generalization**: to what group of observational units can the results be applied to?
  - If an unbiased method of selection was used and there is no non-response or response bias, we can generalize the results to the target population.
  - If a biased method of selection was used or if non-response or response bias is present, we can only generalize the result to the sample or similar observational units.

1.2 Video Notes: Intro to data and Sampling Methods

Read through Sections 1.1 – 1.3 and 2.1 in the course textbook and watch the course videos prior to coming to class. Fill in the following questions to aid in your understanding of the material.

1.2.1 Course Videos

- Data_Basics
- Sampling

Video: Data Basics (sections 1.2.1 and 1.2.2)

Data: observations used to answer research questions

Observational unit or case: the people or things we ______________________________ data from; in a data set, each ________________________ represents one observational unit.

- The number of observational units in a data set is called the ______________________

__________________________ and is denoted by the letter ________________ .

Variable: ________________________________ measured on each observational unit; in a dataset, each ____________________________ represents one variable.

Types of variables

- Categorical variable:

- Quantitative variable:

Example: Review the `loan50` data set discussed in section 1.2.1 of the textbook.

- Identify the sample size. Write it using proper notation.

- What are the observational units?

- Review Table 1.4 in section 1.2.1 of the textbook, listing the 7 variables in the `loan50` data set. Identify which variables are categorical.

- Identify which variables are quantitative.

Summary measure: the *type* of number (mean/proportion/correlation/etc.) used to summarize an entire data set

Examples:

proportion of loans when the person owns a home

mean (or average) interest rate

- The summary measure (mean/proportion/correlation/etc.) and type of plot (bar plot/histogram/scatterplot/etc.) used depends on the type (categorical or quantitative) of variable(s) being summarized!

Optional Notes: Additional Example

The Bureau of Transportation Statistics (*Bureau of Transportation Statistics* 2019) collects data on all forms of public transportation. The data set seen here includes several variables collect on flights departing on a random sample of 150 US airports in December of 2019.

```
airport <- read.csv("data/airport_delay.csv")
glimpse(airport)
#> Rows: 150
#> Columns: 19
#> $ airport            <chr> "ABI", "ABY", "ACV", "ACY", "ADQ", "AEX", "ALB", "~
#> $ city               <chr> "Abilene", "Albany", "Arcata/Eureka", "Atlantic Ci~
#> $ state              <chr> " TX", " GA", " CA", " NJ", " AK", " LA", " NY", "~
#> $ airport_name       <chr> " Abilene Regional", " Southwest Georgia Regional"~
#> $ hub                <chr> "no", "no", "no", "no", "no", "no", "no", "no", "n~
#> $ international      <chr> "no", "no", "no", "yes", "no", "yes", "yes", "yes"~
#> $ elevation_1000     <dbl> 1.7906, 0.1932, 0.2223, 0.0748, 0.0787, 0.0881, 0.~
#> $ latitude           <dbl> 32.4, 31.5, 41.0, 39.5, 57.7, 31.3, 42.7, 35.2, 45~
#> $ longitude          <dbl> -99.7, -81.2, -124.1, -74.6, -152.5, -92.5, -73.8,~
#> $ arr_flights        <int> 195, 81, 215, 293, 54, 282, 943, 410, 53, 32314, 6~
#> $ perc_delay15       <dbl> 16.410256, 13.580247, 23.255814, 15.358362, 12.962~
#> $ perc_cancelled     <dbl> 0.5128205, 0.0000000, 4.1860465, 0.6825939, 14.814~
#> $ perc_diverted      <dbl> 0.00000000, 0.00000000, 2.32558139, 0.68259386, 0.~
#> $ arr_delay          <int> 1563, 1244, 4763, 2905, 329, 1293, 15127, 9705, 25~
#> $ carrier_delay      <int> 459, 890, 1613, 476, 180, 302, 5627, 2253, 439, 10~
#> $ weather_delay      <int> 21, 43, 549, 124, 1, 58, 2346, 168, 1236, 13331, 2~
#> $ nas_delay          <int> 257, 39, 154, 771, 51, 112, 2096, 616, 746, 45674,~
#> $ security_delay     <int> 0, 0, 0, 25, 0, 0, 44, 0, 0, 375, 0, 83, 0, 23, 0,~
#> $ late_aircraft_delay <int> 826, 272, 2447, 1509, 97, 821, 5014, 6668, 108, 10~
```

- What are the observational units?

- Identify which variables are categorical.

- Identify which variables are quantitative.

Video: Sampling (section 2.1)

The method used to collect data will impact **who the results apply to** and **what type of conclusions** can be drawn.

- Target population: all ____ of interest
- Sample: the group of ____ from which data is collected; a subset of the target population

Parameters and statistics

Statistic: a single number which summarizes data from a ____________________________; also called the **point estimate**

- The value of the statistic will _____________________ from one sample to the next, called the _________________________ _________________________

Parameter: a single number which summarizes data from a _________________________________

- The value of the parameter is ____________________________, but is typically ____________________________
- We estimate the value of the parameter by calculating a ____________________________________

Good vs. bad sampling

GOAL: to have a sample that is _________________________ of the _________________________ on the variable(s) of interest

- If the sample is _____________________________, then it is reasonable to believe the value of the ________________________ will be close to the value of the ______________________________!
- Biased sampling method:

Types of Sampling Bias

- Selection bias:

Example of Selection Bias: Newspaper article from 1936 reported that Landon won the presidential election over Roosevelt based on a poll of 10 million voters. Roosevelt was the actual winner. What was wrong with this poll? Poll was completed using a telephone survey and not all people in 1936 had a telephone. Only a certain subset of the population owned a telephone so this subset was over-represented in the telephone survey. The results of the study, showing that Landon would win, did not represent the target population of all US voters.

- Non-response bias:

- To calculate the non-response rate:

$$\frac{\text{number of people who do not respond}}{\text{total number of people selected for the sample}} \times 100\%$$

- For non-response bias to occur must first select people to participate and then they choose not to.

Example of Non-response bias: A company randomly selects buyers to complete a review of an online purchase but some choose not to respond.

- Response bias:

Example of Response Bias: Police officer pulls you over and asks if you have been drinking. Expect people to say no, whether they have been drinking or not.

- Need to be able to predict how people will respond.

Words of caution:

- Convenience samples: gathering data for those who are easily accessible; online polls

    Selection bias?

    Non-response bias?

    Response bias?

- Random sampling reduces ________________________ bias, but has no impact on _______________________ or ____________________ bias.

- Unbiased sample methods:

    Simple random sample

Optional Notes: Additional Examples

Many high schools moved to partial or fully online schooling in Spring of 2020. Did students who graduated in 2020 tend to have a lower GPA during freshman year of college than the previous class of college freshmen? A nationally representative sample of 1000 college students who were freshmen in AY19-20 and 1000 college students who were freshmen in AY20-21 was taken to answer this question.

- What is the target population?

- What is the sample?

Discuss whether any type(s) of bias would be present for each of the data collection methods described below:

A radio talk show asks people to phone in their views on whether the United States should pay off its debt to the United Nations.

- Selection?
- Non-response?
- Response?

The Wall Street Journal plans to make a prediction for the US presidential election based on a survey of its readers and plans to follow-up to ensure everyone responds.

- Selection?
- Non-response?
- Response?

A police detective interested in determining the extent of drug use by high school students, randomly selects a sample of high school students and interviews each one about any illegal drug use by the student during the past year.

- Selection?
- Non-response?
- Response?

1.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. What are the two types of variables?
2. Purpose of random selection:
3. Types of sampling bias:

1.3 Activity 1: Intro to Data Analysis and Sampling Bias

1.3.1 Learning outcomes

- Identify observational units, variables, and variable types in a statistical study.
- Creating a data set
- Identify biased sampling methods.

1.3.2 Terminology review

Statistics is the study of how best to collect, analyze, and draw conclusions from data. In the next few days of class, we will learn the building blocks of the semester. This week in class you will be introduced to the following terms:

- Observational units or cases
- Variables: categorical or quantitative
- Selection bias
- Response bias
- Non-response bias

For more on these concepts, read Chapter 1 and 2 in the textbook and review the Module 1 Vocabulary Review and Key Topics.

Notes on Observational Units and Variables

Review of data set

In the Fall of 2025 a data set was collected on Stat 216 students in class on day 1. The following instructions were provided to the students:

When creating a data set for use in R it is important to use single words or an underscore between words. Each outcome must be written the same way each time. Make sure to use all lowercase letters when responding to the following questions to have consistency between responses. Do not give units of measure for numerical values within the data set. For `Residency` use in_state or out_state as the two outcomes.

- Major: what is your declared major?
- Residency: do you have in-state or out-of-state residency?
- Num_Credits: how many credits are you taking this semester?
- Dominant_hand: are you left or right-handed?
- Hand_span: what is the width of your dominant hand from the tip of your thumb to the tip of your pinky with your hand spread out measured in cm?
- Grip_dominant: what is the grip strength measured in lbs for your dominant hand?
- Grip_nondominant: what is the grip strength measured in lbs for your non-dominant hand?

The following code reads in this data set and gives a snapshot of the variables collected. In a data set, the columns represent the variables collected on the observational units and there is one row of data for each observational unit.

```
data <- read.csv("data/Fall2025_DataCollection.csv")
glimpse(data)
#> Rows: 875
#> Columns: 7
#> $ Major                    <chr> "Microbiology ", "Psychology ", "Business ", "~
#> $ Residency                <chr> "in state", "In state", "Out of state", "Out o~
#> $ Num_Credits              <chr> "15", "19", "17", "16", "15", "18", "16", "15"~
#> $ Dominant_hand            <chr> "right", "right", "right", "right", "right", "~
#> $ Hand_Span                <chr> "8in", "17.5 cm", "22.25", "21 cm", "19", "22"~
#> $ GripStrength_DomHand     <chr> "53.7", "54 lbs", "64", "88.3", "44.4", "86.4"~
#> $ GripStrength_NonDomHand <chr> "45.4", "32.4 lbs ", "68", "84.4", "42.2", "70~
```

1. What are the observational units or cases for the data collected?

2. How many observations are reported in the data set? This is the **sample size**.

3. Using the glimpse of the data, fill in the following table identifying the type of each variable.

- If the variable is categorical, indicate in the third column of the following table whether the variable is binary.
- If the variable is quantitative, indicate in the forth column the units of measure used.

Column	Type of Variable	Binary?	Units?
Major			
Residency			
Num Credits			
Dominant hand			
Hand Span			
Grip strength dominant hand			
Grip strength non-dominant hand			

4. Review the completed data set with your class. Remember that when creating a data set for use in R it is important to use single words or an underscore between words. Each outcome must be written the same way each time to have consistency between responses. Do not include units of measure in the data set when reporting numerical values. Write down some issues found with the created class data set.

Notes on Sampling Methods and Types of bias

Types of bias

Complete Q5 together as a class:

5. A television station is interested in predicting whether or not local voters will pass a referendum to legalize marijuana for adult use. The TV station asks its viewers to phone in and indicate whether they are in favor or opposed to the referendum. Of the 2241 viewers who phoned in, forty-five percent were opposed to legalizing marijuana.

   Sample size:

   Observational units sampled:

   Target population:

   Justify why there is selection bias in this study.

6. To determine if the proportion of out-of-state undergraduate students at Montana State University has increased in the last 10 years, a statistics instructor sent an email survey to 500 randomly selected current undergraduate students. One of the questions on the survey asked whether they had in-state or out-of-state residency. She only received 378 responses.

   Sample size:

   Observational units sampled:

   Target population:

   Justify why there is non-response bias in this study.

7. To gauge the interest of Bozeman City Voters in a new swimming pool, a local organization stood outside of the Bogert Pool in Bozeman, MT, during open hours. One of the questions they asked was, “Since the Bogert Pool is in such bad repair, don’t you agree that the city should fund a new pool?”

   Sample size:

   Observational units sampled:

   Target population:

   Justify why there is response bias in this study.

   Justify why there is selection bias in this study.

1.3.3 Take-home messages

1. There are two types of variables: categorical (groups) and quantitative (numerical measures).
2. We will learn more about summarizing variables later in the semester. Categorical variables are summarized by calculating a proportion from the data and quantitative variables are summarized by finding the mean and the standard deviation (more later on this).
3. There are three types of bias to be aware of when designing a sampling method: selection bias, non-response bias, and response bias.

1.3.4 Additional notes

For more practice on types of bias, as well as identifying the observational units and variable(s) in a study, use the Module 1 Review worksheet in the Unit 1 Review Materials (pgs.109–110).

Use this space to summarize your thoughts and take additional notes on today’s activity and material covered, and to write down the names and contact information of your teammates.

1.4 Activity 2: American Indian Address

1.4.1 Learning outcomes

- Explain why a sampling method is unbiased or biased.
- Identify biased sampling methods.
- Explain the purpose of random selection and its effect on generalization.

1.4.2 Terminology review

In this activity, we will examine unbiased and biased methods of sampling. Some terms covered in this activity are:

- Random sample
- Unbiased vs biased methods of selection
- Generalization

To review these concepts, see Chapter 2 in the textbook.

1.4.3 Class Preparation

Prior to the next class, complete questions 1–3.

American Indian Address

For this activity, you will read a speech given by Jim Becenti, a member of the Navajo American Indian tribe, who spoke about the employment problems his people faced at an Office of Indian Affairs meeting in Phoenix, Arizona, on January 30, 1947 (Moquin and Van Doren 1973). His speech is below:

It is hard for us to go outside the reservation where we meet strangers. I have been off the reservation ever since I was sixteen. Today I am sorry I quit the Santa Fe [Railroad]. I worked for them in 1912–13. You are enjoying life, liberty, and happiness on the soil the American Indian had, so it is your responsibility to give us a hand, brother. Take us out of distress. I have never been to vocational school. I have very little education. I look at the white man who is a skilled laborer. When I was a young man I worked for a man in Gallup as a carpenter's helper. He treated me as his own brother. I used his tools. Then he took his tools and gave me a list of tools I should buy and I started carpentering just from what I had seen. We have no alphabetical language.

We see things with our eyes and can always remember it. I urge that we help my people to progress in skilled labor as well as common labor. The hope of my people is to change our ways and means in certain directions, so they can help you someday as taxpayers. If not, as you are going now, you will be burdened the rest of your life. The hope of my people is that you will continue to help so that we will be all over the United States and have a hand with you, and give us a brotherly hand so we will be happy as you are. Our reservation is awful small. We did not know the capacity of the range until the white man come and say "you raise too much sheep, got to go somewhere else," resulting in reduction to a skeleton where the Indians can't make a living on it. For eighty years we have been confused by the general public, and what is the condition of the Navajo today? Starvation! We are starving for education. Education is the main thing and the only thing that is going to make us able to compete with you great men here talking to us.

By eye selection

1. Circle ten words in Jim Becenti's speech which are a representative sample of the length of words in the entire text. Describe your method for selecting this sample.

2. Fill in the table below with your selected words from the previous question and the length of each word (number of letters/digits in the word):

Observation	Word	Length
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

3. Calculate the mean (average) word length in your selected sample. Is this value a parameter or a statistic?

Notes on sampling

1.4.4 Class Activity

1. Report your mean word length from question 3 to your instructor. Your instructor will create a visualization of the distribution of results generated by your class. Draw a picture of the plot here. Include a descriptive x-axis label. Report the mean and standard deviation of the sample mean word lengths.

The plot created in question 1 is a sampling distribution of statistics. This sampling distribution plots the mean word length from many samples taken from the population of words.

2. The true mean word length of the population of all 359 words in the speech is 3.95 letters. Is this value a parameter or a statistic?

   Where does the value of 3.95 fall in the plot given? Near the center of the distribution? In the tails of the distribution?

3. Would you say the sampling method used ("by-eye" selection) by the class is biased or unbiased? Justify your answer.

4. If the sampling method is biased, what type of sampling bias (selection, response, non-response) is present? What is the direction of the bias, i.e., does the method tend to overestimate or underestimate the population mean word length?

Random selection

Suppose instead of attempting to select a representative sample by eye (which did not work), each student used a random number generator to select a simple random sample of 10 words. A **simple random sample** relies on a random mechanism to choose a sample, without replacement, from the population, such that every sample of size 10 is equally likely to be chosen.

Later in the semester we will learn more about how to use R - for today we will have a short introduction using the `R` studio server. Download the provided R script file from canvas.

5. Login to the R studio server using your netID login. Upload the file to the server using the following instructions.

- Click "Upload" in the "Files" tab in the bottom right window of RStudio. In the pop-up window, click "Choose File", and navigate to the folder where the Activity R script file is saved (most likely in your downloads folder). Click "Open"; then click "Ok".
- Once the file is uploaded, you should see the uploaded file appear in the list of files in the bottom right window. Click on the R script file name to open the file in the Editor window (upper left window).

The following code will take a random sample of 10 words from the population of 359 words.

- In the R script file, highlight and run lines 1–7

```
sample(words$Word, 10)
```

Fill in the table with the random words selected and calculate the word length (number of letters/digits in the word):

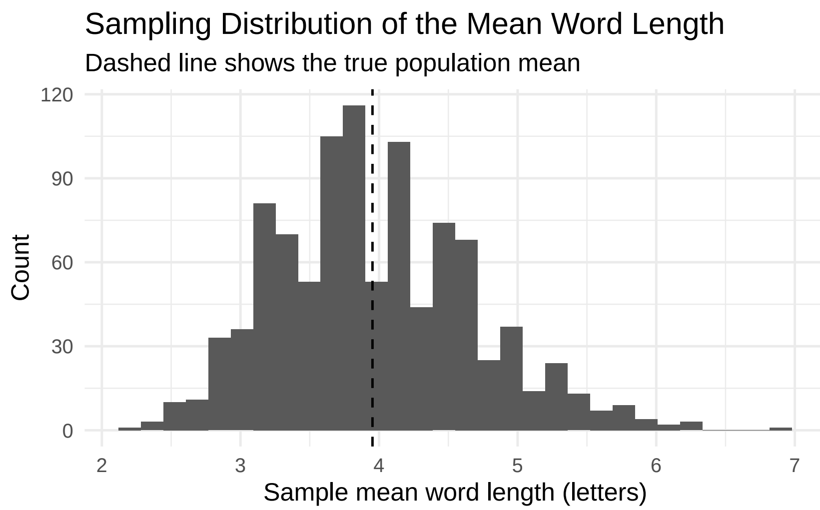
Word	Length

6. Calculate the mean word length in your selected sample above. Is this value a parameter or a statistic?

7. Report your mean word length as instructed. Your instructor will create a visualization of the distribution of results generated by your class. Draw a picture of the plot here. Include a descriptive x-axis label. Report the mean and standard deviation of the samples.

One set of randomly generated sample mean word lengths from a single class may not be large enough to visualize the distribution results.

The following is a plot of 1000 random samples of size 10 created in R. To re-create this plot run lines 11–28 in the R script file.

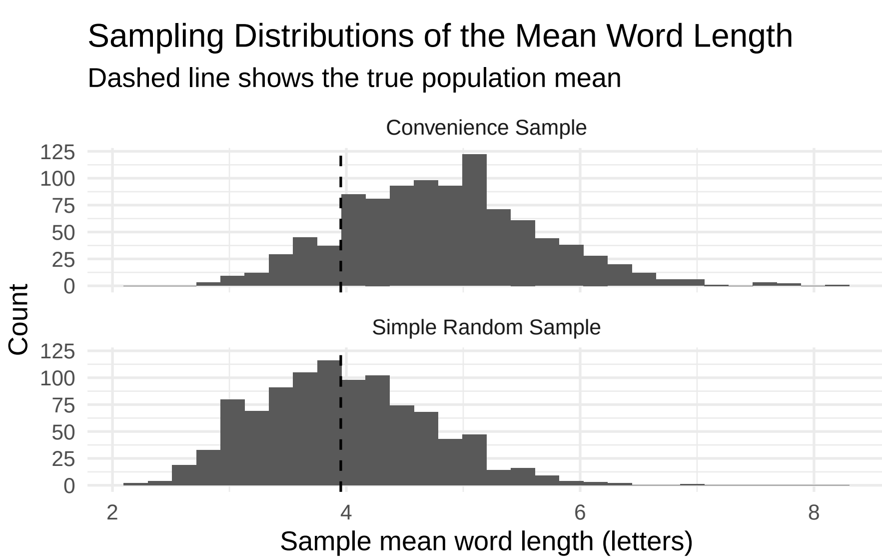


It would be useful to know what the center (mean) is for this sampling distribution. The distribution you created in the R script file may vary a little from what is reported in your coursepack. To find the mean and standard deviation from your distribution run lines 32–33.

```
mean(srs_means)
#> [1] 3.9714
sd(srs_means)
#> [1] 0.7223599
```

8. Would you say this sampling method used (random selection) is biased or unbiased? Justify your answer.

The following two plots show the sampling distribution of convenience samples of size 10 from the population of word length (top) and the simple random sample created above (bottom).



The mean and standard deviation for the convenience sample is provided below.

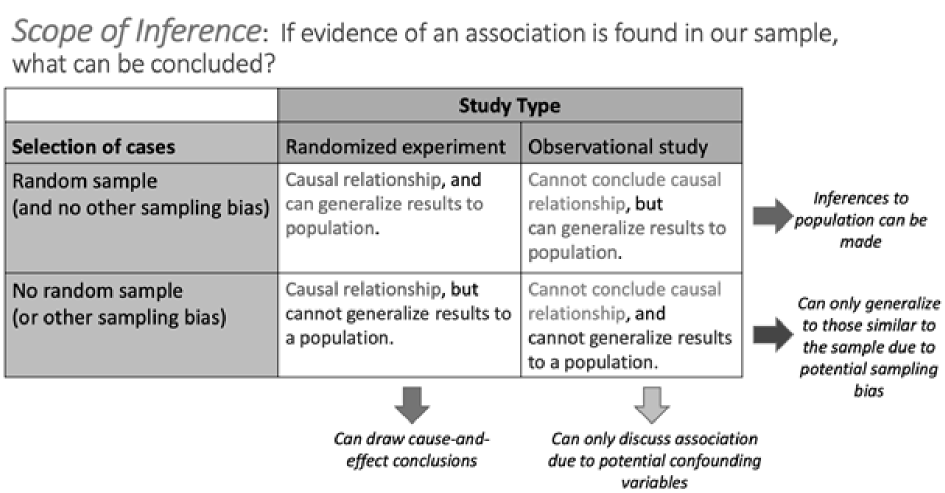
```
mean(convenience_means)
#> [1] 4.8243
sd(convenience_means)
#> [1] 0.8334471
```

9. Compare the two distributions shown above. What is similar between the two distributions? What is different?

10. Using the plots above, explain why using a random number generator to generate a sample is a "better" method than choosing 10 words "by eye" (convenience sample).

11. Is random selection an unbiased method of selection? Explain your answer. Be sure to reference the created plots.

Later in the semester we will learn about the Scope of Inference for a study. A diagram of Scope of Inference is show below:



In this activity we have focused on sampling methods. Note that the sampling method used to select the sample determines the group of observation units the results of a study apply to. This is "generalization." We will explore causation in Module 8.

12. What is the purpose of random selection of a sample from the population?

1.4.5 Take-home messages

1. When we use a biased method of selection, we will over or underestimate the parameter, on average.
2. If the sampling method is biased, inferences made about the population based on a sample estimate will not be valid.
3. Random selection is an unbiased method of selection.
4. To determine if a sampling method is biased or unbiased, we compare the distribution of the estimates to the true value. We want our estimate to be on target or unbiased. When using unbiased methods of selection, the mean of the distribution matches or is very similar to our true parameter.
5. Random selection eliminates selection bias. However, random selection will not eliminate response or non-response bias.

1.4.6 Additional notes

For more practice on sampling methods, use the Module 1 Review worksheet in the Unit 1 Review Materials (pgs.109–110).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

MODULE 2

Probability

2.1 Vocabulary Review and Key Topics

2.1.1 Key topics

Module 2 introduces the concept of probability as a long-run relative frequency and demonstrates how to use hypothetical two-way tables to set up a probability problem and solve for unconditional and conditional probabilities.

2.1.2 Vocabulary

- **Probability** (of an event): the long-run proportion of times the event would occur if the random process were repeated indefinitely (under identical conditions).
- **Conditional probability** (of an event *given* another event): probability of an event calculated dependent on another event having occurred.
- **Probability notation**:
  - $P(A)$: the probability of event A.
    - This is the probability of a single event, *unconditional* probability calculated out of the overall population.
  - $P(A^C)$: the probability of the **complement** of event A, or "A complement".
    - This is the probability of the opposite of event A, or "not A".
    - $P(A^C) = 1 - P(A)$
  - $P(A \text{ and } B)$: the probability of event A and B.
    - The is the probability of an "and" event, *unconditional* probability calculated out of the overall population.
  - $P(A|B)$: the probability of event A given (conditional on) event B.
    - This is a *conditional* probability calculated out of the total population for which event B occurred.

2.2 Video Notes: Probability

Read Chapter 23 in the course textbook. Use the following videos to complete the video notes for Module 2.

2.2.1 Course Videos

- Probability

Video: Probability (chapter 23)

Example: Two variables were collected on a random sample of people who had ever been married; whether a person had ever smoked and whether a person had ever been divorced. The data are displayed in the following table. This survey was based on a random sample in the United States in the early 1990s, so the data should be representative of the adult population who had ever been married at that time.

- Let event D be a person has gone through a divorce
- Let event S be a person smokes

	Has divorced	Has never divorced	Total
Smokes	238	247	485
Does not smoke	374	810	1184
Total	612	1057	1669

- What is the approximate probability that the person smoked?

- What is the approximate probability that the person had ever been divorced?

- Given that the person had been divorced, what is the probability that he or she smoked?

- Given that the person smoked, what is the probability that he or she had been divorced?

- Random process: one in which the outcome is ________________________________
  - Example: Rolling a fair die
- Event: something that could occur, something we want to find the probability of; Denoted using a capital letter.
  - Example: Getting a four when rolling a fair die

- Complement: opposite of the event; Denoted using a ______________________________
  - Example: Getting any value but a four when rolling a fair die
- The probability of an event is the ___________________________ ___________________________ of times the event would occur if the random process were repeated ____________________________; Denoted using $P(event)$
  - Example: the probability of getting a four when rolling a fair die is _________________.
- Unconditional probability: calculated from the _________________________________
  - Unconditional probabilities do **not** depend any event occurring.
  - The denominator of the probability will always be ____.
  - Two types of unconditional probabilities:
    - The probability of a single event
      - Example from section 23.2: The probability of being late to class
    - An "And" probability
      - Example from section 23.2: The probability of finding an open parking spot in Lot 6 **and** being late to class
- Conditional probabilities: calculated ____________________________________ on the occurrence of another event.
  - The denominator of the probability will be the total for the event _________________________ (or the event that is known to have occurred)
  - Example from section 23.2: The probability of being late to class, *given* that you parked in Lot 18
  - Example from section 23.2: The probability of having parked in Lot 18, *given* that you were late to class

Optional practice: For each of the four probability questions about divorce and smoking status from the beginning of these vide notes, determine if the calculated probability was a conditional or unconditional probability and write it using proper notation.

Creating a hypothetical two-way table

Steps:

- Start with a large number like 100000.
- Then use the unconditional probabilities to fill in the row or column totals.
- Now use the conditional probabilities to begin filling in the interior cells.

- Use subtraction to find the remaining interior cells.
- Add the column values together for each row to find the row totals.
- Add the row values together for each column to find the column totals.

Example: As a student at Montana State University, suppose your first class on Mondays is in Wilson Hall at 8:00am and you commute to school. You have a Bobcat parking permit. From past experience, you know that there is a 20% chance of finding an open parking spot in Lot 6 by Animal Bioscience. Otherwise, you have to park in Lot 18 by graduate housing. If you find a spot in Lot 6, you only have a 5% chance of being late to class. However, if you have to park in Lot 18, you have a 15% chance of being late to class. What is the probability that you will be late to class this Monday?

- Let event A = finding an open parking spot in Lot 6 by Animal Bioscience
- Let event C = being late to class

Start by identifying the probability notation for each value given.

- 0.20 =
- 0.05 =
- 0.15 =
- Note for the table below, we will use 100,000 hypothetical Mondays (instead of 1,000 as shown in the textbook)

	C	C^C	Total
A			
A^C			
Total			100,000

For each of the following, determine if the probability is conditional or unconditional. Then write the probability in proper notation and calculate the value.

- What is the probability you are not late to class?
- What is the probability that you park in Lot 6 and you are not late to class

- Given that you were late to class, what is the probability you parked in Lot 18?

Diagnostic tests

- Sensitivity:

- Specificity:

- Prevalence:

Optional Notes: Additional Example

An airline has noticed that 30% of passengers pre-pay for checked bags at the time the ticket is purchased. The no-show rate among customers that pre-pay for checked bags is 5%, compared to 15% among customers that do not pre-pay for checked bags.

- Let event B = customer pre-pays for checked bag
- Let event N = customer no shows

Start by identifying the probability notation for each value given.

- 0.30 =

- 0.05 =

- 0.15 =

	B	B^C	Total
N			
N^C			
Total			100,000

- What is the probability that a randomly selected customer who shows for the flight, pre-purchased checked bags?

2.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. In the parking example, calculate and interpret the following: $P(A^C|C^C) =$.

2. What is the probability notation for 0.15 in the airline example?

2.3 Activity 3: Probability Studies

2.3.1 Learning outcomes

- Recognize and simulate probabilities as long-run frequencies.
- Construct two-way tables to evaluate conditional probabilities.

2.3.2 Terminology review

In today's activity, we will cover two-way tables and probability. Some terms covered in this activity are:

- Proportions
- Probability
- Conditional probability
- Two-way tables

To review these concepts, see Chapter 23 in the textbook.

Notes on probability

The probability of an event is the long-run proportion of times the event would occur if the random process were repeated indefinitely (under identical conditions).

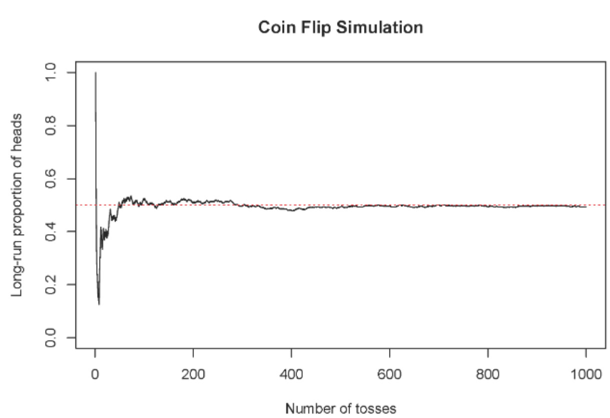
To calculate the probability of an event happening:

$$\text{probability} = \frac{\text{number of ways an event can happen}}{\text{total number of possible outcomes}}$$

For example, to calculate the probability of a coin flip landing on heads; there are only two outcomes (heads or tails) and only one possible way to land on heads.

$$P(heads) = \frac{1}{2} = 0.5$$

The figure below shows the long-run proportion of times a simulated coin flip lands on heads on the y-axis, and the number of tosses on the x-axis. Notice how the long-run proportion starts converging to 0.5 as the number of tosses increases.



In today's activity we will discuss the probability of a single event, the probability of an "and" event, and the probability of a conditional event.

Probability notation

We will use the notation P(event) to represent the probability of an event and use letters to represent events. The following are notations for different probabilities where we are discussing event A and event B:

- $P(A)$ represents the probability of event A
- $P(A^C)$ represents the probability of the complement of event A
  - Additionally, we can find the complement of event A by subtracting the probability of A from 1: $P(A^C) = 1 - P(A)$
- $P(A \text{ and } B)$ represents the probability of events A and B
- $P(A|B)$ represents the probability of event A, given event B
- $P(B|A)$ represents the probability of event B, given event A

Probability questions

For the beginning of this activity we will start with discussing the probabilities associated with drawing a card from a standard card deck. In a card deck there are:

- 52 cards
- Half are red, half are black
- Four suits: spades, hearts, diamonds, and clubs
- Each suit has 13 cards: cards 2–10, ace, jack, queen, and king
- Let A represent the event that a card is an ace
- Let B represent the event that a card is red

To find the probability of selecting an ace, first start with determining how many aces are possible (four) and how many cards will we select from (total of 52).

Find the probability of selecting a card that is not an ace. This is the complement of event A.

Find the probability of selecting a red ace.

Find the probability of selecting an ace given that the card is red.

If a card drawn is an ace, what is the probability the card drawn is red.

Calculating probabilities from a two-way table

Let's start with creating the two-way table. Your instructor will work through creating the table with the class.

1. In 2014, the website FiveThirtyEight examined the works of Bob Ross to see what trends could be found. They determined that of all the paintings he created, 95% of them contained at least one "happy tree." Of those works with a happy tree, 43% contained at least one "almighty mountain." Of the paintings that did not have at least one happy tree, only 10% contained at least one almighty mountain.

Let A = Bob Ross painting contains a happy tree, and B = Bob Ross painting contains an almighty mountain

Create a hypothetical two-way table to represent the situation.

	A	A^C	Total
B			
B^C			
Total			100,000

a. What is the probability that a randomly selected Bob Ross painting contains both a "happy tree" and an "almighty mountain"? Use appropriate probability notation.

b. What is the probability that a selected Bob Ross painting without an "almighty mountain" contains a "happy tree." Use appropriate probability notation.

c. What is the probability that a selected Bob Ross painting does not contain a "happy tree" given it does not contain an "almighty mountain". Use appropriate probability notation.

2. Since the early 1980s, the rapid antigen detection test (RADT) of group A *streptococci* has been used to detect strep throat. A study of the accuracy of this test shows that the **sensitivity**, the probability of a positive RADT given the person has strep throat, is 86% in children, while the **specificity**, the probability of a negative RADT given the person does not have strep throat, is 92% in children. The **prevalence**, the probability of having group A strep, is 37% in children. (Stewart et al. 2014)

   Let $A =$ the event the child has strep throat, and $B =$ the event the child has a positive RADT.

   a. Identify what each numerical value given in the problem represents in probability notation.

   $0.86 =$

   $0.92 =$

   $0.37 =$

   b. Create a hypothetical two-way table to represent the situation.

	A	A^C	Total
B			
B^C			
Total			100,000

   c. Find $P(B)$. What does this probability represent in the context of the problem?

   d. Find the probability that a child with a positive RADT actually has strep throat. What is the notation used for this probability?

   e. What is the probability that a child does not have strep given that they have a positive RADT? What is the notation used for this probability?

2.3.3 Take home messages

1. Conditional probabilities are calculated dependent on a second variable. In probability notation, the variable following | is the variable on which we are conditioning. The denominator used to calculate the probability will be the total for the variable on which we are conditioning.

2. When creating a two-way table we typically want to put the explanatory variable on the columns of the table and the response variable on the rows.

3. To fill in the two-way table, always start with the unconditional variable in the total row or column and then use the conditional probabilities to fill in the interior cells.

2.3.4 Additional notes

To understand probability you need lots of practice! Use the Module 2 Review in the Unit 1 Review Material (pgs. 111–112) for more practice problems.

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

MODULE 3

Exploring Categorical Data: Exploratory Data Analysis and Inference using Simulation-based Methods

3.1 Vocabulary Review and Key Topics

Review the Golden Ticket posted in the resources at the end of the coursepack for a summary of a single categorical variable.

3.1.1 Key topics

Module 3 introduces the steps of the statistical investigation process. We will conduct **exploratory data analysis** (summary statistics and plots) and simulation-based **inference** (hypothesis testing and confidence intervals) in the single categorical variable (one proportion) scenario.

- Notation for a sample proportion: $\hat{p}$
- Notation for a population proportion: π
- Types of plots for a single categorical variable:
  - Frequency bar plot
  - Relative frequency bar plot

Exploratory data analysis is step 3 of the statistical investigation process. We will then use simulation-based methods **to find evidence of an effect by finding a p-value** and **estimating how large the effect is by creating a confidence interval** in the one proportion (one categorical variable) scenario. These are steps 4 and 5 from the steps of the statistical investigation process.

Steps of the statistical investigation process

As we move through the semester we will work through the six steps of the statistical investigation process.

1. Ask a research question.
2. Design a study and collect data.
3. Summarize and visualize the data.
4. Use statistical analysis methods to draw inferences from the data.
5. Communicate the results and answer the research question.
6. Revisit and look forward.

3.1.2 Vocabulary

- **Summary measure**: a numerical quantity that summarizes data. Summary measures covered in STAT 216 include: single proportion, difference in proportions, single mean, paired mean difference, difference in means, correlation, and slope of a regression line.
  - For a single categorical variable, a proportion is calculated.

- **Summary statistic (point estimate)**: the value of a numerical summary measure computed from *sample* data.
  - To interpret in context include:
    - Summary measure (in context)
    - Value of the statistic
- **Parameter of interest**: a numerical summary measure of the entire *population* in which we are interested.
  - The value of the parameter of interest is unknown (unless we have access to the entire population).
  - To write in context:
    - Population word (true, long-run, population)
    - Summary measure (depends on the type of data)
    - Context
      - Observational units
      - Variable(s)
- For a single categorical variable, the category that we are counting the proportion of is generically called a "**success**", with categories not a success labeled "**failure**". Thus, a sample proportion is the "proportion of successes" in the sample: the total number of successes divided by the sample size (n).

Plotting one categorical variable

- **Frequency bar plot**: plots the count (frequency) of observational units in each level of a categorical variable. R code to create a frequency bar plot:

```
object %>% # Data set piped into...
ggplot(aes(x = variable)) +   # This specifies the variable
geom_bar(stat = "count") +  # Tell it to make a bar plot
labs(title = "Don't forget to title your plot!",
   # Give your plot a title
   x = "x-axis label",    # Label the x axis
   y = "Frequency")  # Label the y axis
```

- **Relative frequency bar plot**: plots the proportion (relative frequency) of observational units in each level of a categorical variable. R code to create a relative frequency bar plot:

```
object %>% # Data set piped into...
ggplot(aes(x = variable)) +   # This specifies the variable
geom_bar(aes(y = after_stat(prop), group = 1)) +  # Tell it to make a bar plot with proportions
labs(title = "Don't forget to title your plot!",
   # Give your plot a title
   x = "x-axis label",    # Label the x axis
   y = "Relative Frequency")  # Label the y axis
```

Inference

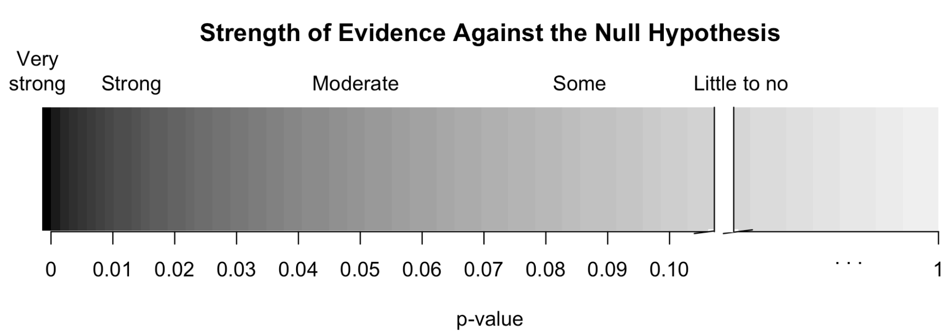
- **Sampling distribution** (of a statistic): the distribution of possible values of a statistic across repeated samples of the same size and under the same conditions.
  - We can create a *simulated* sampling distribution using simulation-based methods to simulate many samples, or we can mathematically model the sampling distribution (theory-based methods).

- **Hypothesis testing**: a formal statistical technique for evaluating two competing possibilities about a population: the null hypothesis and alternative hypothesis.
  - When we observe an effect in a sample, we would like to determine if this observed effect represents an actual effect in the population, or whether it was simply due to random chance.
  - A hypothesis test helps us answer the following question about the population: How strong is the *evidence* of an effect?
- **Null hypothesis**: typically represents a statement of "no difference", "no effect", or the status quo.
  - The null hypothesis is what we assume is true when calculating the p-value. Thus, we can never have evidence *for* the null hypothesis—we cannot "accept" a null hypothesis—we can only find evidence *against* the null hypothesis if the observed data is very unlikely to have occurred under the assumption that the null hypothesis is true.
- **Alternative hypothesis**: represents an alternative claim under consideration and is often represented by a range of possible values for the parameter of interest.
  - The alternative hypothesis is determined by the research question.
- **Hypotheses in notation for a single proportion**: In the hypotheses below, π_0 is the **null value**.

$$H_0 : \pi = \pi_0$$

$$H_A : \pi \left\{ \begin{array}{c} < \\ \neq \\ < \end{array} \right\} \pi_0$$

- **P-value**: the probability of the value of the observed sample statistic or a value more extreme, if the null hypothesis were true.
  - To write in context include:
    * Statement about probability or proportion of samples
    * Statistic (summary measure and value)
    * Direction of the alternative
    * Null hypothesis (in context)
- **Strength of evidence**: the p-value indicates the amount of evidence there is against the null hypothesis. The smaller the p-value the more evidence there is against the null hypothesis.



- **Conclusion** (to a hypothesis test): answers the research question. How much evidence is there in support of the alternative hypothesis?
  - To write in context include:
    * Amount of evidence
    * Parameter of interest
    * Direction of the alternative hypothesis
- **Confidence interval**: an interval estimate for the parameter of interest; an interval of *plausible values* for the parameter.
  - A confidence interval helps us answer the following question about the population: How *large* is the effect?
  - To write in context include:
    * How confident you are (e.g., 90%, 95%, 98%, 99%)
    * Parameter of interest
    * Calculated interval

Simulation-based inference for a single proportion

- **Conditions necessary to use simulation-based methods for inference for a single categorical variable**:
  - **Independence**: observational units must be independent of one another; the outcome of one observational unit should have no influence on the outcome of another.
- **Null distribution**: a sampling distribution of simulated sample statistics created under the assumption that the null hypothesis is true
- **Simulation-based methods to create the null distribution**: a process of using a computer program (e.g., R) to simulate many samples that we would expect based on the null hypothesis.

  R code to use simulation methods for one categorical variable to find the p-value, `one_proportion_test` (from the `catstats` package), is shown below.

```
one_proportion_test(probability_success = xx, # Null hypothesis value
      sample_size = xx, # Enter sample size
      number_repetitions = 10000, # Enter number of simulations
      as_extreme_as = xx, # Observed statistic
      direction = "xx", # Specify direction of alternative hypothesis
      summary_measure = "proportion") # Reporting proportion or number of successes?
```

- **Bootstrapping**: creating a simulated sample of the same size as the original sample by sampling with replacement from the original sample.
- **Simulation-based methods to create the bootstrap distribution**: a process of using a computer program to simulate many bootstrapped samples.

  R code to use simulation methods for one categorical variable to find a confidence interval, `one_proportion_bootstrap_CI` (from the `catstats` package), is shown below.

```
one_proportion_bootstrap_CI(sample_size = xx, # Sample size
                number_successes = xx, # Observed number of successes
                number_repetitions = 10000, # Number of bootstrap samples to use
                confidence_level = 0.95) # Confidence level as a decimal
```

- **Percentile method**: process to find the confidence interval from the bootstrap distribution.
  - A 90% confidence interval will be found between the 5th and 95th percentiles of the bootstrap distribution.
  - A 95% confidence interval will be found between the 2.5th and 97.5th percentiles of the bootstrap distribution.
  - A 99% confidence interval will be found between the 0.5th and 99.5th percentiles of the bootstrap distribution.

3.2 Video Notes: Analysis of One Categorical Variable

Read Chapter 3, 4, 9, 10 and Sections 14.1 and 14.2 in the course textbook. Use the following videos to complete the video notes for Module 3.

3.2.1 Course Videos

- One_Categorical_Variable_EDA
- Hypothesis_Testing
- Simulation_Tests_One_Categorical_Variable
- Confidence_Intervals
- Bootstrap_Intervals_One_Categorical_Variable

Video: One Categorical Variable EDA (sections 4.1 and 4.2 - summarizing and displaying one categorical variable)

- A ____________________ is calculated on data from a sample
- The _______________________ is calculated on data from a population; it is typically unknown and therefore we will define it *in words*
- The parameter should include:
  - Population word (true, long-run, population)
  - Summary measure (depends on the type of data)
  - Context
    - Observational units
    - Variable(s)

Categorical data can be numerically summarized by calculating a _____________________ from the data set.

Notation used for the population proportion:

Notation used for the sample proportion:

Categorical data can be summarized in a ____________________________ table, which reports counts or a ______________________ ______________________ table, which reports proportions.

Types of plots for a single categorical variable:

Example from section 4.1 using the `email` data set:

- Identify the observational units.
- What type of variable is `number`?

- Is Table 4.3 a frequency or relative frequency table?

- What proportion of emails had no number in them? Is this value a statistic or a parameter? Write it using proper notation.

- Compare the two plots (frequency bar plot and relative frequency bar plot) of the variable `number` in Figure 4.1 in section 4.2 of the text. Note that the x-axis is the ______________________ in the two plots. However, the _____________________ differs. The scale for the frequency bar plot goes from ____ and the scale for the relative frequency bar plot is from ____.

Optional Notes: Additional Example

Gallatin Valley is the fastest growing county in Montana. You'll often hear Bozeman residents complaining about the 'out-of-staters' moving in. A local real estate agent recorded data on a random sample of 100 home sales over the last year at her company and noted where the buyers were moving from as well as the age of the person or average age of a couple buying a home. The variable age was binned into two categories, "Under30" and "Over30." Additionally, the variable, state the buyers were moving from, was created as a binary variable, "Out" for a location out of state and "In" for a location in state.

The following code reads in the data set, `moving_to_mt` and names the object moving.

```
moving <- read.csv("data/moving_to_mt.csv")
```

The `R` function `glimpse` was used to give the following output.

```
glimpse(moving)
```

```
#> Rows: 100
#> Columns: 4
#> $ From      <chr> "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA", ~
#> $ Age_Group <chr> "Under30", "Under30", "Under30", "Under30", "Under30", "Unde~
#> $ Age       <int> 25, 26, 27, 27, 29, 29, 35, 37, 49, 63, 65, 77, 22, 24, 24, ~
#> $ InOut     <chr> "Out", "Out", "Out", "Out", "Out", "Out", "Out", "Out", "Out~
```

- What are the observational units in this study?

- What type of variable is `Age`?

- What type of variable is `Age_Group`?

To further analyze the categorical variable, `From`, we can create either a frequency table:

```
#>   From  n
#> 1   CA 12
#> 2   CO  8
#> 3   MT 61
#> 4   WA 19
```

Or a relative frequency table:

```
#>   From  n freq
#> 1   CA 12 0.12
#> 2   CO  8 0.08
#> 3   MT 61 0.61
#> 4   WA 19 0.19
```

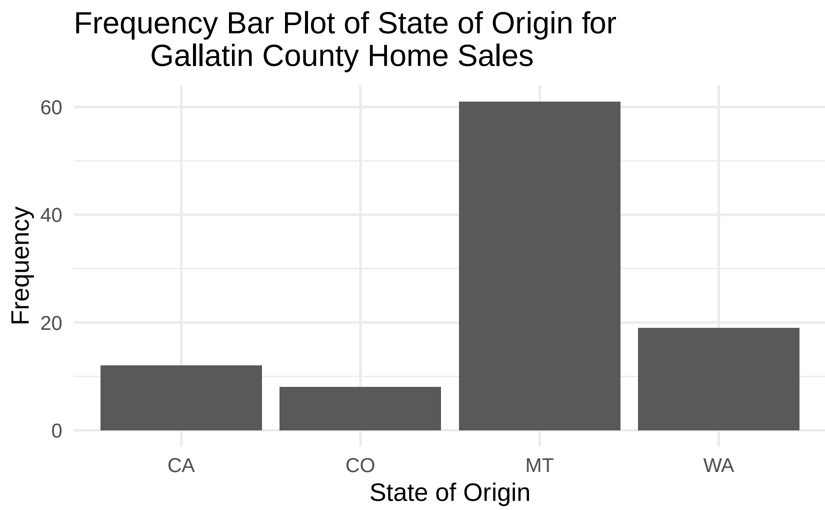
- How many home sales have buyers from WA?

- What proportion of sampled home sales have buyers from WA?

- What notation is used for the proportion of home sale buyers that that are from WA?

The following code in `R` will create a frequency bar plot of the variable, `From`.

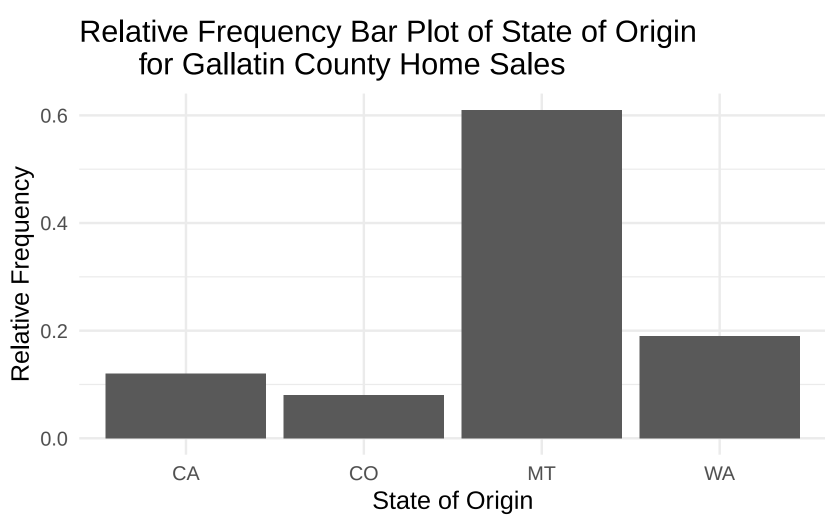
```
moving %>%
    ggplot(aes(x = From))+ #Enter the variable to plot
    geom_bar(stat = "count") +
    labs(title = "Frequency Bar Plot of State of Origin for
         Gallatin County Home Sales",
         #Title your plot (type of plot, observational units, variable)
       y = "Frequency", #y-axis label
       x = "State of Origin") #x-axis label
```



- What can we see from this plot?

Additionally, we can create a relative frequency bar plot.

```
moving %>%
  ggplot(aes(x = From))+ #Enter the variable to plot
  geom_bar(aes(y = after_stat(prop), group = 1)) +
  labs(title = "Relative Frequency Bar Plot of State of Origin
       for Gallatin County Home Sales",
       #Title your plot
       y = "Relative Frequency", #y-axis label
       x = "State of Origin") #x-axis label
```



Video: Hypothesis Testing (chapter 9)

Purpose of a hypothesis test:

- Use data collected on a sample to give information about the population.
- Determines ____________________________ of _________________________________ of an effect

General steps of a hypothesis test

1. Write a research question and hypotheses.
2. Collect data and calculate a summary statistic.
3. Model a sampling distribution which assumes the null hypothesis is true.
4. Calculate a p-value.
5. Draw conclusions based on a p-value.

Hypotheses

- Always hypothesize about the ____________________________, so will always be written in terms of ____________________________.
- Both null and alternative hypotheses will compare the parameter to the same value, only the *sign* will change.

Null hypothesis

- Skeptical perspective, no difference, no effect, random chance
- What the researcher hopes is ________________________.
- Will always use the ____________________________ sign.

Notation:

Alternative hypothesis

- New perspective, a chance, a difference, an effect
- What the researcher hopes is _________________________.
- The sign of the alternative hypothesis reflects the ___________________________ ____________________________.

Notation *(write all three options)*:

Simulation vs. Theory-based Methods

Simulation-based method

Condition: Observational units must be **independent**, meaning

Creation of the null distribution

- Simulate many samples assuming

- Find the proportion of ______________________________ at least as extreme as the observed ______________________________

- The null distribution estimates the sample to sample variability expected in the population

Theory-based method

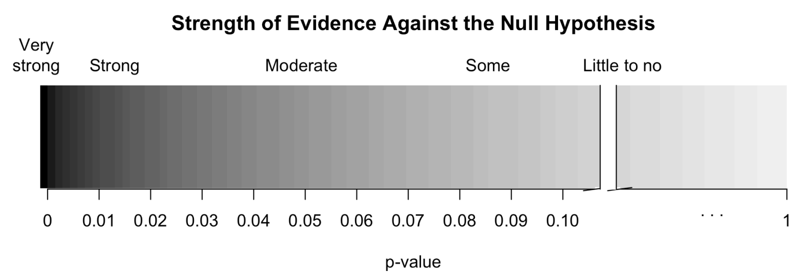
Conditions: Independence and a large enough sample size

- Use a mathematical model to determine a distribution under the null hypothesis
- Compare the observed sample statistic to the model to calculate a probability
- *Theory-based methods will be discussed in the next module*

P-value

- What does the p-value measure?
  - Probability of observing the ______________________________ or ___________________________ assuming the ___________________ hypothesis is ______________________.

- How much evidence does the p-value provide against the null hypothesis?



- The ____________________________ the p-value, the ______________________________ the evidence **against** the null hypothesis.

- Write a conclusion based on the p-value.

  - Answers the _________________________ question.

  - Amount of ___________________________ in support of the ___________________________ hypothesis.

- Decision: can we reject or fail to reject the null hypothesis?

  - Significance level: cut-off of "small" vs "large" p-value

    - p-value $\leq \alpha$

      - Strong enough evidence against the null hypothesis

      - Decision:

      - Results are ________________________________ significant.

    - p-value $> \alpha$

      - Not enough evidence against the null hypothesis

      - Decision:

      - Results are not _____________________________ significant.

Hypothesis Testing/Justice System

- Null hypothesis: The defendant is _____________________________
- Alternative hypothesis: The defendant is ________________________________
- Prosecution collects evidence (data) and presents that evidence to a jury
- Two possible outcomes:
  - Strong evidence against ______________________ -> jury finds the defendant ________________________
  - Not enough evidence against __________________________ -> jury finds the defendant __________________________

Example: Martian Alphabet

Review the Martian Alphabet example from sections 9.1 and 9.3.2

- Write the null hypothesis in words and in notation:

  In words: H_0 :

  In notation: $H_0 : \pi =$ _______, where π represents the _______________ proportion of humans that would _______________________________________.

- Write the alternative hypothesis in words and in notation:

In words: H_A :

In notation: $H_A : \pi$ ______________ (a ____________ sign represents the research question - that humans have a ______________________________ for choosing Bumba on the left)

- What is the value of the statistic? Write it using proper notation.

- Review Figure 9.2. Why is the plot centered around 0.5? What does each "dot" on the plot represent?

- Does the statistic seem likely or unlikely to occur if humans are picking a shape randomly? What about Figure 9.2 helps us to answer this question?

- The p-value of the test is _________________________. This p-value provides _________________________ evidence against the ______________________________ hypothesis and for the ______________________________ hypothesis. The results from this study ______________________________ be considered statistically significant.

Video: Simulation Tests for One Categorical Variable (section 14.1)

Review the Medical Consultant example from section 14.1.

Identify the observational units.

Identify the variable collected.

Is the variable categorical or quantitative? If categorical, define a "success"; if quantitative, state the units of measure.

Define the parameter in words and write it using proper notation

Write the null and alternative hypotheses in words and in proper notation:

In words:

H_0 :

H_A :

In notation:

H_0 :

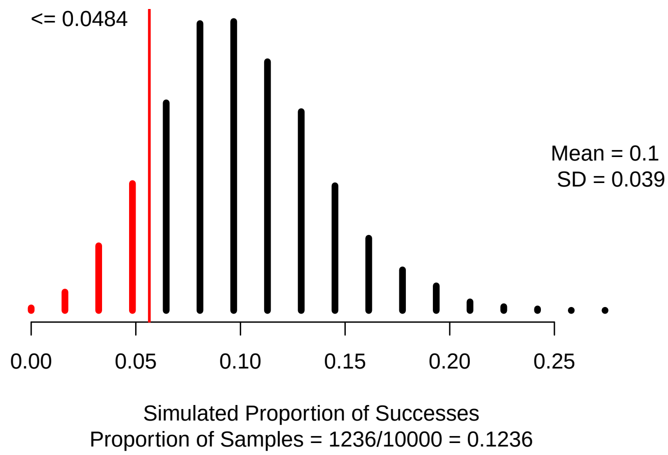
H_A :

Calculate the value of the statistic and write it using proper notation:

Simulation-based method

- Simulate many samples assuming $H_0 : \pi = \pi_0$
  - Create a spinner with that represents the null value
  - Spin the spinner n times
  - Calculate and plot the simulated sample proportion from each simulation
  - Repeat 10000 times (simulations) to create the null distribution
  - Find the proportion of simulations at least as extreme as $\hat{p}$

```
set.seed(216)
one_proportion_test(probability_success = 0.10, # Null hypothesis value
          sample_size = 62, # Enter sample size
          number_repetitions = 10000, # Enter number of simulations
          as_extreme_as = 0.0484, # Observed statistic
          direction = "less", # Specify direction of alternative hypothesis
          summary_measure = "proportion") # Reporting proportion or number of successes?
```



Explain why the null distribution is centered at the value of approximately 0.10:

Where is the red line plotted on the null distribution? Why is that value important?

Why is the area left of the red line shaded on the null distribution?

What is the p-value of the test?

Interpretation of the p-value:

- Statement about probability or proportion of samples
- Statistic (summary measure and value) and Direction of the alternative
- Null hypothesis (population reference, summary measure, equal to null value)
- Context of the problem (observational units, variable (if variable is categorical, define "success"))

Two example p-value interpretations are provided below. Identify each of the components within the two examples:

- Example 1: There is a 12.36% probability of observing three (3) or fewer complications out of sixty-two (63) liver donor surgeries in the US, if the true proportion of complications is 0.10.
- Example 2: If 10% of all liver donor surgeries in the US result in a complication, 1236/10000 samples would simulate a proportion of complications of 0.048 or less.

Conclusion:

- Amount of evidence
- For the alternative hypothesis (population reference, summary measure, direction, null value)
- Context (observational units, variable (if variable is categorical, define "success"))

An example conclusion is provided below. Identify each of the components within the example:

- There is little to no that the proportion of all liver donor surgeries in the US that result in a complication is less than 0.10.

Generalization:

- Can the results of the study be generalized to the target population?

Video: Confidence intervals (chapter 10)

statistic $\pm$ margin of error

Vocabulary:

- Point estimate:

- Margin of error:

Purpose of a confidence interval

- To give an ________________________________ ______________________________ for the parameter of interest
- Determines how ______________________ an effect is

Sampling distribution

- Ideally, we would take many samples of the same _________________ from the same population to create a sampling distribution
- But only have 1 sample, so we will ___________________________ with ___________________________ from the one sample.
- Need to estimate the sampling distribution to see the ___________________________ in the sample

Simulation-based methods

Bootstrap distribution:

- Write the response variable values on cards
- Sample with replacement n times (bootstrapping)
- Calculate and plot the simulated difference in sample means from each simulation
- Repeat 10000 times (simulations) to create the bootstrap distribution
- Find the cut-offs for the middle X% (confidence level) in a bootstrap distribution.

What is bootstrapping?

- Assume the "population" is many, many copies of the original sample.
- Randomly sample with replacement from the original sample n times.

Example: Medical Consultant

Review the Medical Consultant example in section 10.1.

- Is 0.048 the statistic or the parameter? How do you know?

- Review Figure 10.6 in section 10.1.2.
  - Why is the plot centered near 0.048? What does each "dot" on the plot represent?

  - Why are the 2.5th and 97.5th percentiles of the bootstrap distribution marked? What are the values of these two percentiles?

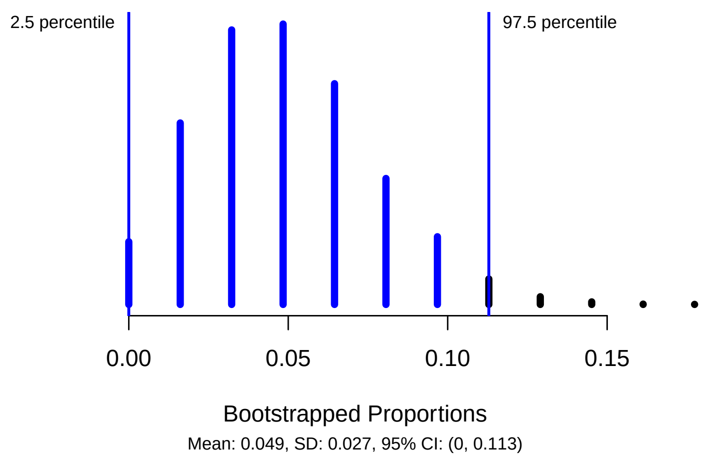
- Interpret the confidence interval: We are __________% ______________________________ that, in the ______________________, the true probability (or proportion) of complications is between ________________ and ____________.

Video: Bootstrap Confidence Intervals for One Categorical Variable (section 14.2)

Review the Medical Consultant study from section 14.2.

Bootstrap distribution:

```
set.seed(216)
one_proportion_bootstrap_CI(sample_size = 62, # Sample size
                    number_successes = 3, # Observed number of successes
                    number_repetitions = 10000, # Number of bootstrap samples to use
                    confidence_level = 0.95) # Confidence level as a decimal
```



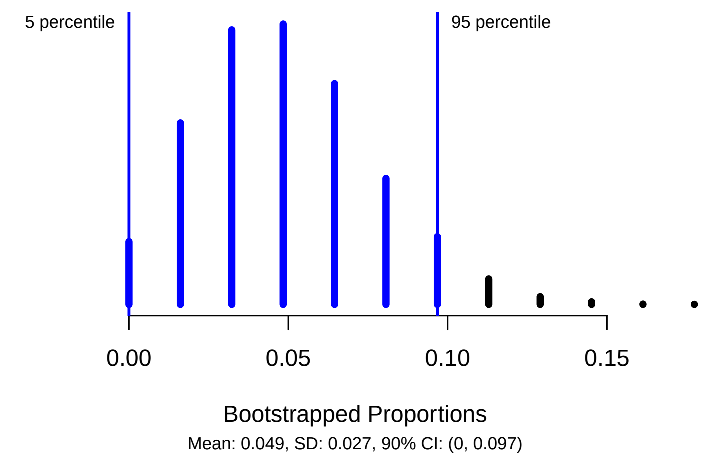
Confidence interval interpretation:

- How confident you are (e.g., 90%, 95%, 98%, 99%)
- Parameter of interest (including context: observational units, variable (if variable is categorical, define "success"))
- Calculated interval

How does changing the confidence level impact the width of the confidence interval?

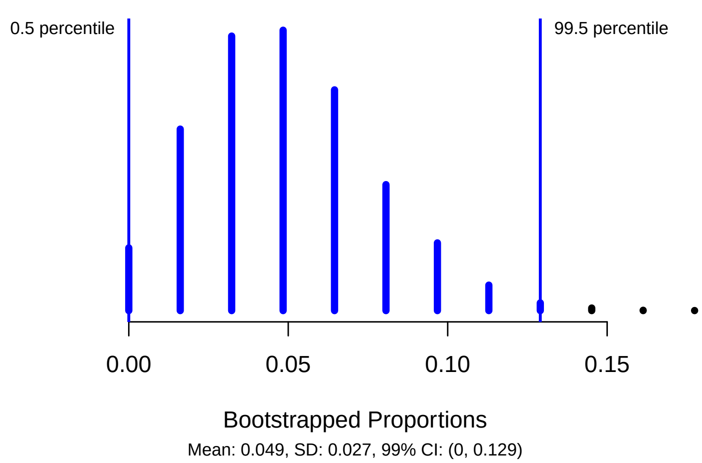
90% Confidence Interval:

```
set.seed(216)
one_proportion_bootstrap_CI(sample_size = 62, # Sample size
                    number_successes = 3, # Observed number of successes
                    number_repetitions = 10000, # Number of bootstrap samples to use
                    confidence_level = 0.90) # Confidence level as a decimal
```



99% Confidence Interval:

```
set.seed(216)
one_proportion_bootstrap_CI(sample_size = 62, # Sample size
                    number_successes = 3, # Observed number of successes
                    number_repetitions = 10000, # Number of bootstrap samples to use
                    confidence_level = 0.99) # Confidence level as a decimal
```



3.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. What is the summary measure calculated from a single categorical variable?

2. Write the alternative hypothesis for the Martian Alphabet study in notation? How was the direction of the alternative hypothesis determined?

3. Do the results of the confidence interval for the Medical Consultant study *match* the results based on the p-value?

3.3 Activity 4: Helper-Hinderer Part 1 — Simulation-based Hypothesis Test

3.3.1 Learning outcomes

- Identify the two possible explanations (one assuming the null hypothesis and one assuming the alternative hypothesis) for a relationship seen in sample data.
- Given a research question involving a single categorical variable, construct the null and alternative hypotheses in words and using appropriate statistical symbols.
- Describe and perform a simulation-based hypothesis test for a single proportion.

3.3.2 Terminology review

In today's activity, we will work through a simulation-based hypothesis testing for a single categorical variable. Some terms covered in this activity are:

- Parameter of interest
- Null hypothesis
- Alternative hypothesis
- Simulation

To review these concepts, see Chapters 9 & 14 in your textbook.

3.3.3 Steps of the statistical investigation process

We will work through a five-step process to complete a hypothesis test for a single proportion.

- **Ask a research question** that can be addressed by collecting data. What are the researchers trying to show?
- **Design a study and collect data**. This step involves selecting the people or objects to be studied and how to gather relevant data on them.
- **Summarize and visualize the data**. Calculate summary statistics and create graphical plots that best represent the research question.
- **Use statistical analysis methods to draw inferences from the data**. Choose a statistical inference method appropriate for the data and identify the p-value and/or confidence interval after checking assumptions. In this study, we will focus on using randomization to generate a simulated p-value.
- **Communicate the results and answer the research question**. Using the p-value and confidence interval from the analysis, determine whether the data provide statistical evidence against the null hypothesis. Write a conclusion that addresses the research question.

Notes on one categorical variable

3.3.4 Helper-Hinderer

A study by Hamblin, Wynn, and Bloom reported in Nature (Hamblin et al. 2007) was intended to check young kids' feelings about helpful and non-helpful behavior. Non-verbal infants ages 6 to 10 months were shown short videos with different shapes either helping or hindering the climber. As a class we will watch this short video to see how the experiment was run: https://youtu.be/anCaGBsBOxM. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy? In the study, of the 16 infants age 6 to 10 months, 14 chose the *helper* toy and 2 chose the *hinderer* toy.

- Observational units:
- Variable:
  - Success:

Ask a research question

- Research question:

Design a study and collect data

Before using statistical inference methods, we must check that the cases are independent. The sample observations are independent if the outcome of one observation does not influence the outcome of another. One way this condition is met is if data come from a simple random sample of the target population.

1. Are the cases independent? Justify your answer.

R Instructions

For almost all activities and labs it will be necessary to upload the provided R script file from Canvas for that day. Your instructor will highlight a few steps in uploading files to and using RStudio.

The following are the steps to upload the necessary R script file for this activity:

- Download the Activity R script file from Canvas.

- Click “Upload” in the “Files” tab in the bottom right window of RStudio. In the pop-up window, click “Choose File”, and navigate to the folder where the Activity R script file is saved (most likely in your downloads folder). Click “Open”; then click “Ok”.
- You should see the uploaded file appear in the list of files in the bottom right window. Click on the file name to open the file in the Editor window (upper left window).

Notice that the first three lines of code contain a prompt called `library`. Packages needed to run functions in R are stored in directories called libraries. When using the MSU RStudio server, all the packages needed for the class are already installed. We simply must tell R which packages we need for each R script file. We use the prompt `library` to load each **package** (or library) needed for each activity. Note, these `library` lines MUST be run each time you open a R script file in order for the functions in R to work.

- Highlight and run lines 1–3 to load the packages needed for this activity. Notice the use of the # symbol in the R script file. This symbol is not part of the R code. It is used by these authors to add comments to the R code and explain what each call is telling the program to do.

R will ignore everything after a # symbol when executing the code. Refer to the instructions following the # symbol to understand what you need to enter in the code.

```
library(tidyverse)
library(ggplot2)
library(catstats)
```

Throughout activities, we will often include the R code you would use in order to produce output or plots. These “code chunks” appear in gray. In the code chunk below, we demonstrate how to read the data set into R using the `read.csv()` function. The line of code shown below (line 7 in the R script file) reads in the data set and names the data set `infants`.

Summarize and visualize the data

The following code reads in the data set and gives the number of infants in each level of the variable, whether the infant chose the helper or the hinderer.

- Highlight and run lines 7 and 8 to check that you get the same counts as shown below

```
 # Read in data set
infants <- read.csv("https://math.montana.edu/courses/s216/data/infantchoice.csv")
infants %>% count(choice)  # Count number in each choice category
```

```
#>     choice  n
#> 1   helper 14
#> 2 hinderer  2
```

The following formula is used to calculate the proportion of successes in the sample.

$$\hat{p} = \frac{\text{number of successes}}{\text{total number of observational units}}$$

2. Using the R output and the formula given, calculate the summary statistic (sample proportion) to represent the research question. Recall that `choosing the helper toy` is a considered a success. Use appropriate notation.

To visually display this data we can use either a frequency bar plot or a relative frequency bar plot.

- Enter the variable name `choice` for `variable` in the R code to create the frequency bar plot.
- Note the name of the title is given in line 16 and includes the **type of plot**, **observational units**, and **variable name**.
- Highlight and run lines 13–19 to create the plot

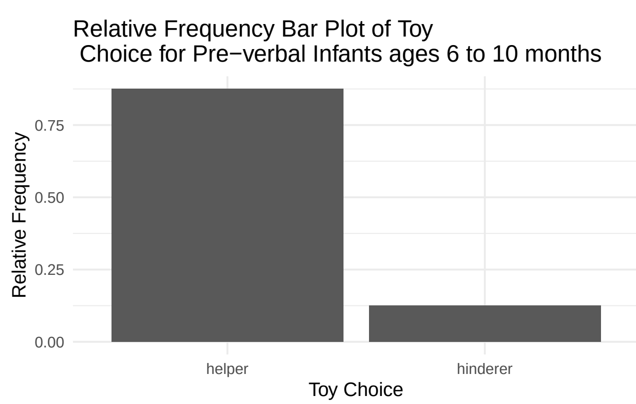
```
infants %>% # Data set piped into...
    ggplot(aes(x = variable)) +   # This specifies the variable
    geom_bar(stat = "count") +  # Tell it to make a bar plot
    labs(title = "Frequency Bar Plot of Toy Choice for Pre-verbal Infants ages 6 to 10 months",
       # Give your plot a title
       x = "Toy Choice",   # Label the x axis
       y = "Frequency")  # Label the y axis
```

3. Sketch the frequency bar plot created below.

We could also choose to display the data as a proportion in a **relative frequency** bar plot. To find the relative frequency, the count in each level of `choice` is divided by the sample size. This calculation is the sample proportion for each level of `choice`. Notice that in the following code we told R to create a bar plot with proportions.

- In the R script file, highlight and run lines 23–29 to create the relative frequency bar plot.

```
infants %>% # Data set piped into...
    ggplot(aes(x = choice)) +   # This specifies the variable
    geom_bar(aes(y = after_stat(prop), group = 1)) +  # Tell it to make a bar plot with proportions
    labs(title = "Relative Frequency Bar Plot of Toy \n Choice for Pre-verbal Infants ages 6 to 10 months
       # Give your plot a title
       x = "Toy Choice",   # Label the x axis
       y = "Relative Frequency")  # Label the y axis
```



4. Which features in the relative frequency bar plot are the same as the frequency bar plot? Which are different?

We cannot assess whether infants are more likely to choose the helper toy based on the statistic and plot alone. The next step is to analyze the data by using a hypothesis test to discover if there is evidence against the null hypothesis.

Use statistical analysis methods to draw inferences from the data

Notes on hypotheses tests

When performing a hypothesis test, we must first identify the null hypothesis. The null hypothesis is written about the parameter of interest, or the value that summarizes the variable in the population.

The parameter of interest is a statement about what we want to find about the population. The following must be included when writing the parameter of interest.

- Population word (true, long-run, population)
- Summary measure (depends on the type of data)
- Context
  - Observational units
  - Variable(s)

Parameter of interest:

If the children are just randomly choosing the toy, we would expect half (0.5) of the infants to choose the helper toy. This is the null value for our study.

Null Hypothesis (in words):

The notation used for a population proportion (or probability, or true proportion) is π. Since this summarizes a population, it is a parameter. When writing the **null hypothesis** in notation, we set the parameter equal to the null value, $H_0 : \pi = \pi_0$.

Null Hypothesis (in notation):

The **alternative hypothesis** is the claim to be tested and the direction of the claim (less than, greater than, or not equal to) is based on the research question.

- Based on the research question from question 1, are we testing that the parameter is greater than 0.5, less than 0.5 or different than 0.5?

Alternative hypothesis (in words):

Alternative hypothesis (in notation):

Remember that when utilizing a hypothesis test, we are evaluating two competing possibilities. For this study the **two possibilities** are either…

- The true proportion of infants who choose the helper is 0.5 and our results just occurred by random chance; or,
- The true proportion of infants who choose the helper is greater than 0.5 and our results reflect this.

Notice that these two competing possibilities represent the null and alternative hypotheses.

We will now simulate one sample of a **null distribution** of sample proportions. The null distribution is created under the assumption the null hypothesis is true. In this case, we assume the true proportion of infants who choose the helper is 0.5, so we will create 10000 (or more) different simulations of 16 infants under this assumption.

Let's think about how to use a coin to create one simulation of 16 infants under the assumption the null hypothesis is true. Let heads equal infant chose the helper toy and tails equal infant chose the hinderer toy.

5. How many times would you flip a coin to simulate the sample of infants?

6. Flip a coin 16 times recording the number of times the coin lands on heads. This represents one simulated sample of 16 infants randomly choosing the toy. Calculate the proportion of coin flips that resulted in heads.

7. Is the value from question 6 closer to 0.5, the null value, or closer to the sample proportion, 0.875?

Report the number of coin flips you got as indicated by your instructor.

8. Sketch the graph created by your instructor of each student's proportion of heads out of 16 coin flips.

9. Circle the observed statistic (value from question 2) on the distribution shown above. Where does this statistic fall in this distribution: Is it near the center of the distribution (near 0.5) or in one of the tails of the distribution?

10. Is the observed statistic likely to happen or unlikely to happen if the true proportion of infants who choose the helper is 0.5? Explain your answer using the plot.

In the next class, we will continue to assess the strength of evidence against the null hypothesis by using a computer to simulate 10000 samples when we assume the null hypothesis is true.

3.3.5 Take-home messages

1. Two types of plots are used for plotting categorical variables: frequency bar plots, relative frequency bar plots.
2. In a hypothesis test we have two competing hypotheses, the null hypothesis and the alternative hypothesis. The null hypothesis represents either a skeptical perspective or a perspective of no difference or no effect. The alternative hypothesis represents a new perspective such as the possibility that there has been a change or that there is a treatment effect in an experiment.
3. In a simulation-based test, we create a distribution of possible simulated statistics for our sample if the null hypothesis is true. Then we see if the calculated observed statistic from the data is likely or unlikely to occur when compared to the null distribution.
4. To create one simulated sample on the null distribution for a sample proportion, spin a spinner with probability equal to π_0 (the null value), n times or draw with replacement n times from a deck of cards created to reflect π_0 as the probability of success. Calculate and plot the proportion of successes from the simulated sample.

3.3.6 Additional notes

Activities 4–6 cover the material in Module 3. Use the Module 3 Review in the Unit 1 Review Material (pgs. 113–116) for more practice problems.

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

3.4 Activity 5: Helper-Hinderer (continued)

3.4.1 Learning outcomes

- Describe and perform a simulation-based hypothesis test for a single proportion.
- Interpret and evaluate a p-value for a simulation-based hypothesis test for a single proportion.
- Explore what a p-value represents

3.4.2 Steps of the statistical investigation process

In today's activity we will continue with steps 4 and 5 in the statistical investigation process. We will continue to assess the Helper-Hinderer study from last class.

- **Ask a research question** that can be addressed by collecting data. What are the researchers trying to show?
- **Design a study and collect data**. This step involves selecting the people or objects to be studied and how to gather relevant data on them.
- **Summarize and visualize the data**. Calculate summary statistics and create graphical plots that best represent the research question.
- **Use statistical analysis methods to draw inferences from the data**. Choose a statistical inference method appropriate for the data and identify the p-value and/or confidence interval after checking assumptions. In this study, we will focus on using randomization to generate a simulated p-value.
- **Communicate the results and answer the research question**. Using the p-value and confidence interval from the analysis, determine whether the data provide statistical evidence against the null hypothesis. Write a conclusion that addresses the research question.

3.4.3 Helper-Hinderer

In class today, we will revisit the study on infants as described below.

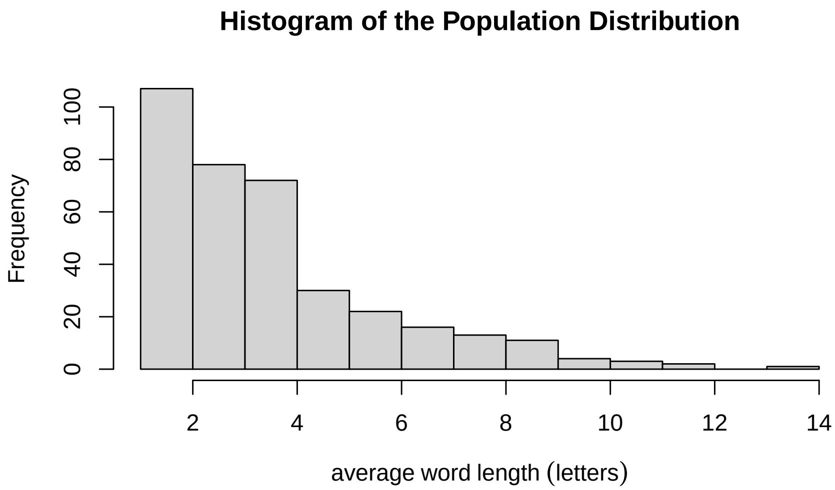
A study by Hamblin, Wynn, and Bloom reported in Nature (Hamblin et al. 2007) was intended to check young kids' feelings about helpful and non-helpful behavior. Non-verbal infants ages 6 to 10 months were shown short videos with different shapes either helping or hindering the climber. As a class we will watch this short video to see how the experiment was run: https://youtu.be/anCaGBsBOxM. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy? In the study, of the 16 infants age 6 to 10 months, 14 chose the *helper* toy and 2 chose the *hinderer* toy.

Sampling distribution of the sample proportion.

The sampling distribution is a distribution of a statistic from all possible samples of size n from the population. The mean of the sampling distribution should be close to the parameter value and the standard deviation of the sampling distribution will estimate the variability of statistics from sample to sample.

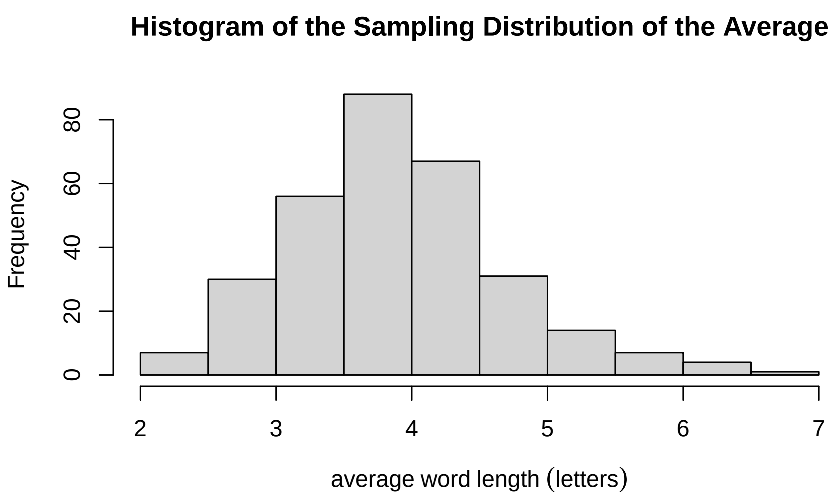
To create a sampling distribution of the statistic, we would take infinite samples of size n from the population and plot the statistic from each sample. To illustrate this idea we will revisit the Becenti address looking at the mean word length.

```
#>   min Q1 median Q3 max     mean       sd   n missing
#> 1   1  2      3  5  14 3.952646 2.300228 359       0
```



Note that the mean of this small population ($N = 359$) is 3.9526. Typically, we don't have such small population sizes so we need a way to simulate the sampling distribution of the statistic to estimate the sample to sample variability. Repeat samples of size 10 are taken from the population of words and the sample mean of each sample is plotted below.

```
#>   min  Q1 median  Q3 max     mean        sd   n missing
#> 1 2.1 3.3    3.9 4.4 6.7 3.934557 0.7954711 305       0
```



The mean of this simulated distribution is 3.934557, which is close to the true mean from the population. The standard deviation is 0.7954711 - this measures how far each statistic is from the true mean word length, on average. Later in the semester, we will learn how this variability is affected by the sample size.

Today, we will use the computer to simulate the sampling distribution of the sample proportion by creating a null distribution of 10000 different samples of 16 infants, plotting the proportion who will choose the helper in each sample, based on the assumption that the true proportion of infants who choose the helper is 0.5 (or that the null hypothesis is true).

To use the computer simulation, we will need to enter the

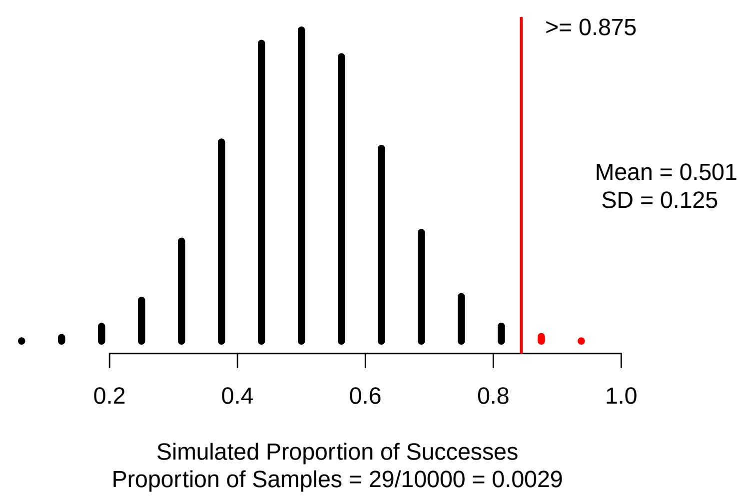
- assumed "probability of success" (π_0),
- "sample size" (the number of observational units or cases in the sample),
- "number of repetitions" (the number of samples to be generated - typically we use 10000),
- "as extreme as" (the observed statistic), and
- the "direction" (matches the direction of the alternative hypothesis).

1. What values should be entered for each of the following into the one proportion test to create 10000 simulations?

- Probability of success (null value):
- Sample size (n):
- Number of repetitions (typically use 10000 simulations):
- As extreme as (value of statistic):
- Direction (`"greater"`, `"less"`, or `"two-sided"`):

The following shows one possible distribution of simulated sample proportions assuming the null hypothesis, true proportion of non-verbal infants aged 6 to 10 months that will choose the helper toy is 0.5, is true.

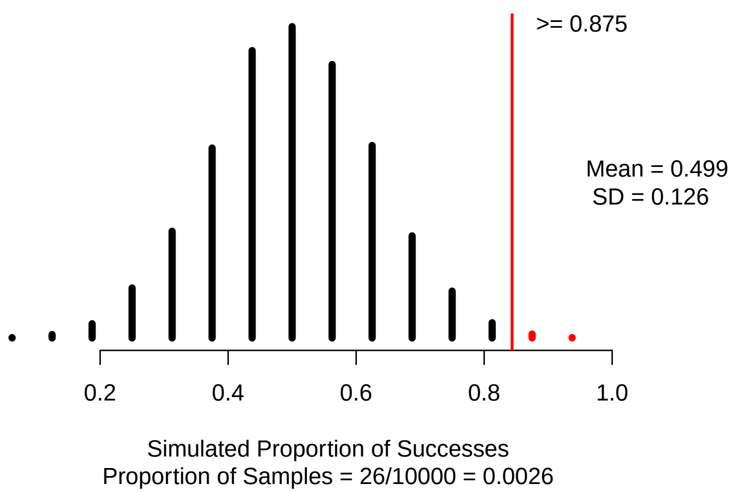
```
set.seed(100)
one_proportion_test(probability_success = 0.5, # Null hypothesis value
          sample_size = 16, # Enter sample size
          number_repetitions = 10000, # Enter number of simulations
          as_extreme_as = 0.875, # Observed statistic
          direction = "greater", # Specify direction of alternative hypothesis
          summary_measure = "proportion") # Reporting proportion or number of successes?
```



Notes on the null distribution

Using R, we will use the `one_proportion_test()` function (in the `catstats` package) to simulate another null distribution of sample proportions and compute a p-value. Using the provided `R` script file, fill in the values/words for each `xx` with your answers from question 1 in the one proportion test to create a null distribution with 10000 simulations. Then highlight and run lines 1–16. *Note that the value for the mean, standard deviation, and p-value changed slightly.*

```
set.seed(216)
one_proportion_test(probability_success = 0.5, # Null hypothesis value
          sample_size = 16, # Enter sample size
          number_repetitions = 10000, # Enter number of simulations
          as_extreme_as = 0.875, # Observed statistic
          direction = "greater", # Specify direction of alternative hypothesis
          summary_measure = "proportion") # Reporting proportion or number of successes?
```



For the remainder of the questions in this activity use this 2nd simulated null distribution.

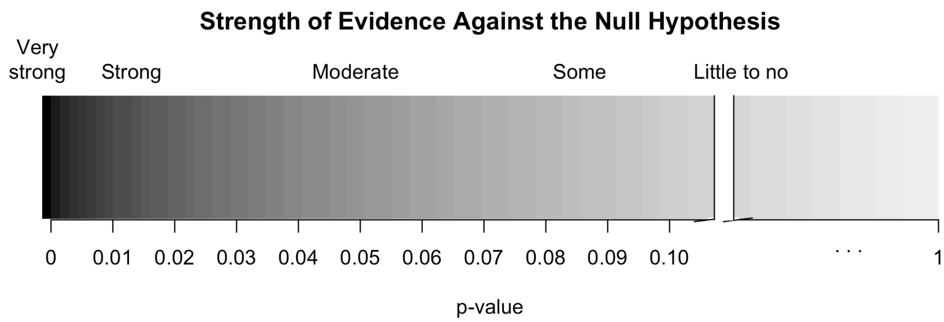
2. Circle the observed statistic (value calculated in Activity 4) on the null distribution. Where does this statistic fall in the null distribution: Is it near the center of the distribution (near 0.5) or in one of the tails of the distribution?

3. Is the observed statistic likely to happen or unlikely to happen if the true proportion of infants who choose the helper is 0.5? Explain your answer using the plot.

4. Using the simulation, what is the proportion of simulated samples that generated a sample proportion at the observed statistic or greater, if the true proportion of infants who choose the helper is 0.5? *Hint*: Look under the simulation.

Notes on the p-value

The value in question 4 is the **p-value**. The smaller the p-value, the more evidence we have against the null hypothesis.



Interpret the p-value

The p-value measures the probability that we observe a sample proportion as extreme as what was seen in the data or more extreme (matching the direction of the H_A) IF the null hypothesis is true. This is a conditional probability, calculated dependent on the null hypothesis being true. Represented in probability notation:

$$P(\text{statistic or more extreme}|\text{the null hypothesis is true})$$

p-value interpretation:

Communicate the results and answer the research question

When we write a conclusion we answer the research question by stating how much evidence there is in support of the alternative hypothesis.

Conclusion:

Generalization

5. To what group of observational units can the results be generalized to?

3.4.4 Take-home messages

1. The null distribution is created based on the assumption the null hypothesis is true. We compare the sample statistic to the distribution to find the likelihood of observing this statistic.
2. The p-value measures the probability of observing the sample statistic or more extreme (in direction of the alternative hypothesis) if the null hypothesis is true.
3. The smaller the p-value of the test, the more evidence there is **against** the null hypothesis.

3.4.5 Additional notes

Activities 4–6 cover the material in Module 3. For more practice with simulation hypothesis testing of a single categorical variable use the Module 3 Review worksheet in the Unit 1 Review Materials (pgs. 113–116).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

3.5 Activity 6: Helper-Hinderer — Simulation-based Confidence Interval

3.5.1 Learning outcomes

- Use bootstrapping to find a confidence interval for a single proportion.
- Interpret a confidence interval for a single proportion.

3.5.2 Terminology review

In today's activity, we will introduce simulation-based confidence intervals for a single proportion. Some terms covered in this activity are:

- Parameter of interest
- Bootstrapping
- Confidence interval

To review these concepts, see Chapters 10 & 14 in your textbook.

3.5.3 Helper-Hinderer

In the last class, we found very strong evidence that the true proportion of infants who will choose the helper character is greater than 0.5. But what *is* the true proportion of infants who will choose the helper character? We will use this same study to estimate this parameter of interest by creating a confidence interval.

As a reminder: A study by Hamblin, Wynn, and Bloom reported in Nature (Hamblin et al. 2007) was intended to check young kids' feelings about helpful and non-helpful behavior. Non-verbal infants ages 6 to 10 months were shown short videos with different shapes either helping or hindering the climber. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy? In the study, of the 16 infants age 6 to 10 months, 14 chose the *helper* toy and 2 chose the *hinderer* toy.

A **point estimate** (our observed statistic) provides a single plausible value for a parameter. However, a point estimate is rarely perfect; usually there is some error in the estimate. In addition to supplying a point estimate of a parameter, a next logical step would be to provide a plausible *range* of values for the parameter. This plausible range of values for the population parameter is called an **interval estimate** or **confidence interval**.

Activity intro

1. What is the value of the point estimate (sample statistic)?

2. If we took another random sample of 16 infants, would we get the exact same point estimate? Explain why or why not.

In today's activity, we will use bootstrapping to find a 95% confidence interval for π, the parameter of interest.

Notes on Confidence Intervals

Use statistical analysis methods to draw inferences from the data

3. Write out the parameter of interest in words, in context of the study. What does π represent?

To create the null distribution we flipped a coin 16 times to simulate infants randomly choosing the helper toy with a probability of 50%.

4. Why can't we use a coin to simulate the bootstrap distribution.

To create the bootstrap distribution.

- First we would label the cards to represent the sample statistic: 14 helper and 2 hinderer.
- Sample with replacement 16 times

5. Using the cards provided by your instructor, create one bootstrap sample. Report your simulated sample proportion on the whiteboard.

To use the computer simulation to create a bootstrap distribution, we will need to enter the

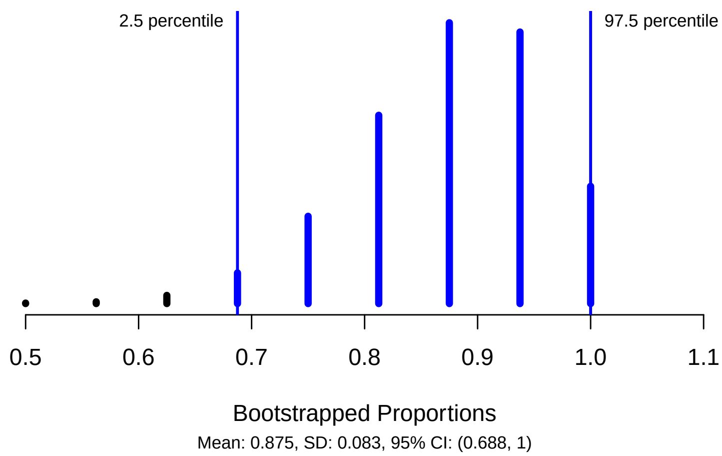
- "sample size" (the number of observational units or cases in the sample),
- "number of successes" (the number of cases that choose the helper character),
- "number of repetitions" (the number of samples to be generated), and
- the "confidence level" (which level of confidence are we using to create the confidence interval).

6. What values should be entered for each of the following into the simulation to create the bootstrap distribution of sample proportions to find a 95% confidence interval?

- Sample size (n):
- Number of successes:
- Number of repetitions (at least 10000):
- Confidence level (as a decimal):

We will use the `one_proportion_bootstrap_CI()` function in R (in the `catstats` package) to simulate the bootstrap distribution of sample proportions and calculate a confidence interval. Using the provided R script file, fill in the values/words for each `xx` with your answers from question 6 in the one proportion bootstrap confidence interval (CI) code to create a bootstrap distribution with 10000 simulations. Then highlight and run lines 1–9.

```
one_proportion_bootstrap_CI(sample_size = 16, # Sample size
                            number_successes = 14, # Observed number of successes
                            number_repetitions = 10000, # Number of bootstrap samples to use
                            confidence_level = 0.95) # Confidence level as a decimal
```



Notes on the bootstrap distribution

95% Confidence Interval:

Interpretation of the 95% confidence interval in context.

Communicate the results and answer the research question

7. Is the value 0.5 (the null value) in the 95% confidence interval?

Explain how this indicates that the p-value provides strong evidence against the null.

Statistical Analysis Write-up

In many of our studies we will write a paragraph summarizing the results of the study. The following is a summary paragraph for the infant study. The paragraph should include an introduction, the interpretation of the sample statistic, p-value and interpretation, confidence interval and interpretation, conclusion statement, and generalization.

An introduction including the observational units and variable(s) in the study and their role and type. The introduction should also include the sampling method and research question.

Researchers were interested if infants observe social cues and would be more likely to choose the helper toy. The observational units in this study are 16 non-verbal infants ages 6 to 10 months old. This is a convenience sample of infants. On each infant the researchers recorded whether or not the infant choose the helper toy. This is a categorical variable. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy?

Report and interpret the sample statistic in context of the study.

From the study, it was reported that the proportion of the 16 infants that chose the helper toy was 0.875.

Report how the p-value was found. Include an interpretation of the p-value.

A simulation null distribution with 10000 simulations was created in RStudio. The p-value was found by calculating the proportion of simulations in the null distribution at the sample statistic of 0.875 and greater. This resulted in a p-value of 0.0026. We would observe a sample proportion of 0.875 or greater with a probability of 0.0026, IF we assume the true proportion of non-verbal infants age 6 to 10 months who would choose the helper toy is 0.5.

Report and interpret the confidence interval.

In addition, a 95% confidence interval was found for the parameter of interest. We are 95% confident that the true proportion of non-verbal infants age 6 to 10 months who will choose the helper toy is between 0.688 and 1.

Write a conclusion for the study.

Based on the p-value of 0.0026, there is very strong evidence that the true proportion of non-verbal infants age 6 to 10 months who will choose the helper toy is greater than 0.5.

And last, report the scope of inference for the study. Since there is only a single categorical variable in this study, we only need to discuss generalization.

The results of this study can be generalized to the sample of non-verbal infants or similar observational units as the researchers did not select a random sample.

3.5.4 Take-home messages

1. The goal in a hypothesis test is to assess the strength of evidence for an effect, while the goal in creating a confidence interval is to determine how large the effect is. A **confidence interval** is a range of *plausible* values for the parameter of interest.

2. In simulation-based methods (bootstrapping), a simulated distribution of possible sample statistics is created showing the possible sample-to-sample variability. Then we find the middle X percent of the distribution around the sample statistic using the percentile method to give the range of values for the confidence interval. This shows us that we are $X\%$ confident that the parameter is within this range, where X represents the level of confidence.

3. When the null value is within the confidence interval, it is a plausible value for the parameter of interest; thus, we would find a larger p-value for a hypothesis test of that null value. Conversely, if the null value is NOT within the confidence interval, we would find a small p-value for the hypothesis test and strong evidence against this null hypothesis.

4. To create one simulated sample on the bootstrap distribution for a sample proportion, label n cards with the original responses. Draw with replacement n times. Calculate and plot the resampled proportion of successes.

3.5.5 Additional notes

Activities 4–6 cover the material in Module 3. For more practice with simulation confidence intervals for a single categorical variable use the Module 3 Review worksheet in the Unit 1 Review Materials (pgs. 113–116).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

MODULE 4

Inference for a Single Categorical Variable: Theory-based Methods

4.1 Vocabulary Review and Key Topics

Review the Golden Ticket posted in the resources at the end of the coursepack for a summary of a single categorical variable.

4.1.1 Key topics

Module 4 introduces theory-based inference methods (hypothesis testing and confidence intervals) for a single categorical variable. We also explore what "confidence level" means and which parts of a study impact the width of a confidence interval and the p-value.

- Theory-based methods should give the same results as simulation-based methods if the sample size is large enough. For a single categorical variable, the sample size is large enough if the success-failure condition is met.
- If repeated samples of the same size are taken from the population, 95% of samples will create a 95% confidence interval that contains the value of the parameter of interest.

4.1.2 Vocabulary

- **Theory-based methods**: when specific conditions are met, the distribution of sample statistics if we were to repeatedly sample from the population can be fit with a theoretical distribution.
- **Conditions for the sampling distribution of $\hat{p}$ to follow an approximate normal distribution**:
  - **Independence**: the sample's observations are independent, e.g., are from a simple random sample. (*Remember*: This also must be true to use simulation-based methods!)
  - **Large enough sample size**: Success-failure condition: we *expect* to see at least 10 successes and 10 failures in the sample, $n\pi \geq 10$ and $n(1-\pi) \geq 10$. Since π is typically unknown, we consider this condition to be met if we observe at least 10 successes and 10 failures in our data set: $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$.
- **Standard normal distribution**: a theoretical distribution that is bell-shaped, centered on the mean of zero, and has a standard deviation of one, denoted in notation by $N(0, 1)$.
- **Standard error of a statistic**: an estimated standard deviation of the statistic as it would vary across repeated samples of the same size under the same conditions.
  - The standard error tells us about how far we would expect an observed sample statistic to fall from the true parameter value for which it is estimating, on average.
- **Standardized statistic**: calculation to standardize the sample statistic in order to compare the standardized value to the theoretical distribution.
  - Calculated by subtracting the null value from the sample statistic, then dividing by the standard error:
$$\frac{\text{statistic} - \text{null value}}{\text{standard error}}$$

  - Measures the number of standard errors the sample statistic is above (if positive) or below (if negative) the null value.

- **Standard error of the sample proportion assuming the null is true**: measures the how far each possible sample proportion is from the true proportion, on average, and is calculated using the null value:

$$SE_0(\hat{p}) = \sqrt{\frac{\pi_0 \times (1 - \pi_0)}{n}}$$

.

- **Standardized sample proportion**: standardized statistic for a single categorical variable calculated using:

$$Z = \frac{\hat{p} - \pi_0}{SE_0(\hat{p})},$$

If the conditions for the sampling distribution of $\hat{p}$ to follow an approximate normal distribution are met, and if the true value of π is equal to the null value of π_0, the standardized sample proportion, Z, will have an approximate *standard* normal distribution.

- The theory-based **p-value** for hypothesis testing involving proportions can be found in R by using the `pnorm` function to find the probability of the observed standardized statistic or one more extreme (in the direction of H_A). This probability is the area under a *standard normal distribution* at or more extreme than the observed standardized statistic.
  - Enter the value of the standardized statistic for `xx`.
  - If a "greater than" alternative, change `lower.tail = TRUE` to `FALSE`.
  - If a two-sided test, multiply by 2.

```
pnorm(xx, lower.tail=TRUE)
```

- **Margin of error**: half the width of the confidence interval. For a single proportion, the margin of error is:

$$ME = z^* \times SE(\hat{p})$$

where z^* is the **multiplier**, corresponding to the desired confidence level found from the standard normal distribution. For example, for a 95% confidence level, the middle 95% of the standard normal distribution falls between $-z^* = -1.96$ and $z^* = 1.96$.

- **Standard error of the sample proportion for a confidence interval** (not assuming the null is true):

$$SE(\hat{p}) = \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}$$

- To find the endpoints of a confidence interval, add and subtract the margin of error to the sample statistic. The confidence interval for a population proportion is:

$$\hat{p} \pm ME$$

- R code to find the **multiplier** for the confidence interval using theory-based methods involving proportions.
  - `qnorm` will give you the multiplier using the standard normal distribution.
  - Enter the percentile for the given level of confidence (e.g., 0.975 for a 95% confidence level).

```
qnorm(percentile, lower.tail=TRUE)
```

4.2 Video Notes: Inference for One Categorical Variable using Theory-based Methods

Read Chapter 11 and Sections 14.3 and 14.4 in the course textbook. Use the following videos to complete the video notes for Module 4.

4.2.1 Course Videos

- Theory_Based_Inference
- Theoretical_Tests_One_Categorical_Variable
- Theoretical_Intervals_One_Categorical_Variable

Theory-based methods

Video: Theory-Based Inference (chapter 11)

The Central Limit Theorem tells us that the ____________________ distribution of a sample proportion (and sample mean and sample differences) will be approximately ____________________ if the sample size is ____________________ ______________________.

The ____________________ of the distribution of sample proportions (sampling distribution) from thousands of samples will be bell-shaped/symmetric (Normal), if the sample size is large enough and the observations are ______________________.

- $\hat{p} \sim N(\pi, \sqrt{\frac{\pi\times(1-\pi)}{n}})$

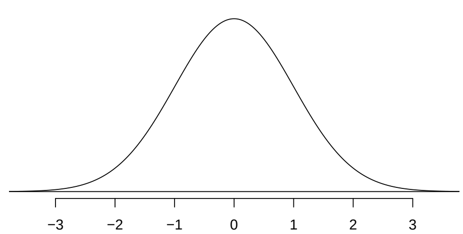
Conditions of the CLT:

- Independence (*also must be met to use simulation methods*): the response for one observational unit will not influence another observational unit

- Large enough sample size:

Normal distribution:

- Bell-shaped and ____________________
- Standard normal distribution: $N(0, 1)$



Standardized statistic: Z - score

- $Z = \frac{\text{statistic} - \text{null value}}{\text{standard error of the statistic}}$

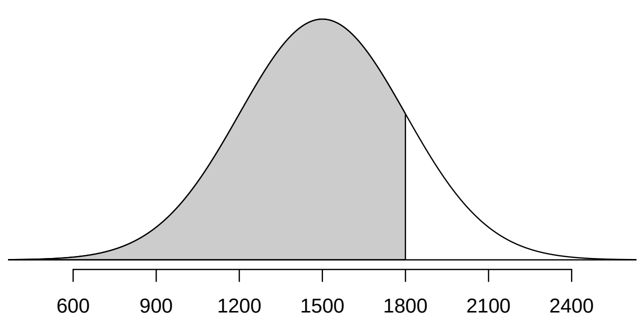
- Measures the number of ______________________ ____________________ the statistic is above or below the null value

Examples from section 11.2: SAT scores follow a nearly normal distribution with a mean of 1500 points and a standard deviation of 300 points. Find and interpret the z-score for Nell, who scored 1800 points on their SAT. Round your answer to three decimal places.

ACT scores also follow a nearly normal distribution with mean of 21 points and a standard deviation of 5 points. Find and interpret the z-score for Sian, who scored 24 points on their ACT. Round your answer to three decimal places. Who performed better - Nell or Sian?

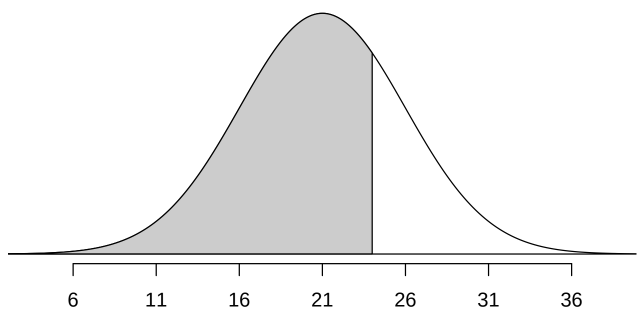
In a Normal curve, the area under the curve is equal to 1, representing a probability. Therefore the shaded area represents the **percentile** for Nell, or the probability of scoring less than or equal to 1800 points on the SAT.

```r
library(openintro)
normTail(m = 1500, s = 300, L = 1800)
pnorm(mean = 1500, sd = 300, q = 1800)
#> [1] 0.8413447
```



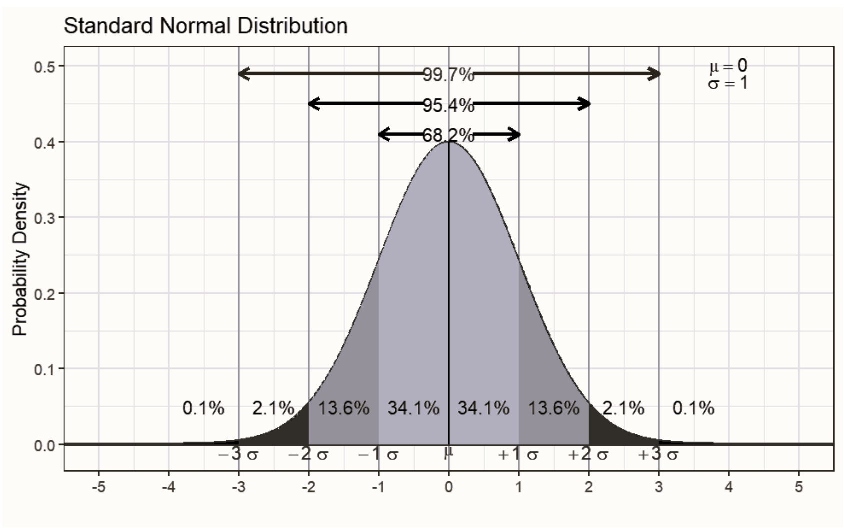
We can also reverse that order. Given a percentage, we can find the associated percentile, or quantile. Here we display calculating the value that cuts off the lower 0.73 proportion of ACT scores using the qnorm() function.

```
qnorm(mean = 21, sd = 5, p = 0.726)
#> [1] 24.0038
normTail(m = 21, s = 5, L = 24)
```



68-95-99.7 Rule

- 68% of Normal distribution within 1 SD of the mean (mean – SD, mean + SD)
- 95% within (mean – 2SD, mean + 2SD)
- 99.7% within (mean – 3SD, mean + 3SD)



General steps of a hypothesis test

1. Write a research question and hypotheses.
2. Collect data and calculate a summary statistic.
3. Model a sampling distribution which assumes the null hypothesis is true.

   - If the conditions are met, the sampling distribution of a sample proportion which assumes the null hypothesis is true is the standard Normal (N(0,1)) distribution

4. Calculate a p-value.

   - Calculate the standardized sample proportion (Z) and compare that to the standard Normal (N(0,1)) distribution.

   - P-value will be the area under the curve beyond Z (in the direction of the alternative hypothesis)

5. Draw conclusions based on a p-value.

Video: Theoretical Tests for One Categorical Variable (section 14.3.1 and 14.3.2)

Review the payday loan borrowers example from section 14.3.2.

Identify the observational units.

Identify the variable collected.

Is the variable categorical or quantitative? If categorical, define a "success"; if quantitative, state the units of measure.

Define the parameter in words and write it using proper notation

Write the null and alternative hypotheses in words and in proper notation:

In words:

H_0 :

H_A :

In notation:

H_0 :

H_A :

Calculate the value of the statistic and write it using proper notation:

Conditions for inference using theory-based methods:

- Independence:
  - The outcome of one observation does not influence the outcome of another.
  - Taking a random sample is one way to satisfy this condition.
- Large enough sample size:

Are the conditions met to analyze the payday loan borrowers data using theory-based methods?

To use theory-based methods to perform a hypothesis test:

- 1st: Calculate the standardized statistic
- 2nd: Find the area under the standard normal distribution at least as extreme as the standardized statistic

Equation for the standard error of the sample proportion, assuming the null hypothesis is true:

- This value measures how far each possible sample statistic is from the null value, on average.
- This should give a similar value to the ____________________ ____________________ of a simulated null distribution!

Equation for the standardized sample proportion (standardized statistic when analyzing one categorical variable):

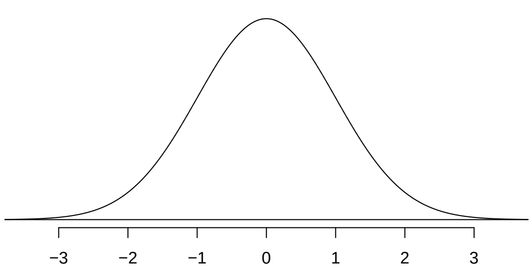
- This value measures how many ________________________ __________________________ the sample ________________________ is above/below the _____________________ _______________________.

Calculate the standard error of the sample proportion of payday loan borrowers that would support the regulation *when assuming the null hypothesis is true*:

Calculate the standardized sample proportion of payday loan borrowers that would support the regulation (the Z-score):

Interpret the standardized statistic

Label the standardized sample proportion (standardized statistic) on the standard Normal distribution below and shade the area representing the p-value.



To find the p-value, find the area under the standard Normal distribution at the standardized statistic and more extreme (in the direction of the alternative hypothesis).

```
pnorm(0.59, lower.tail = FALSE)
```

```
#> [1] 0.2775953
```

Interpretation of the p-value:

- Statement about probability or proportion of samples
- Statistic (summary measure and value) and Direction of the alternative
- Null hypothesis (population reference, summary measure, equal to null value)
- Context of the problem (observational units, variable (if variable is categorical, define "success"))

Conclusion:

- Amount of evidence
- For the alternative hypothesis (population reference, summary measure, direction, null value)
- Context (observational units, variable (if variable is categorical, define "success"))

Decision at a significance level of 0.05 ($\alpha = 0.05$):

Generalization:

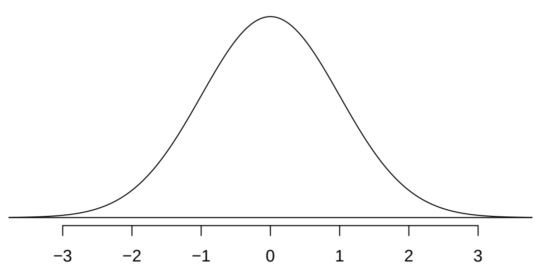
- Can the results of the study be generalized to the target population?

Video: Theoretical Confidence Intervals for One Categorical Variable (section 14.3.3)

- Interval of _______________ values for the parameter of interest
- $CI = \text{statistic} \pm \text{margin of error}$

Theory-based method for a single categorical variable

- $CI = \hat{p} \pm (z^* \times SE(\hat{p}))$
- Multiplier (z^*) is the value at a certain __________________ under the standard normal distribution



For a 95% confidence interval:

```
qnorm(0.975, lower.tail=TRUE)
```

```
#> [1] 1.959964
```

- When creating a confidence interval, we no longer assume the ____________________ hypothesis is true. Use ____________ to calculate the sample to sample variability, rather than π_0.

Equation for the standard error of the sample proportion, *NOT* assuming the null is true:

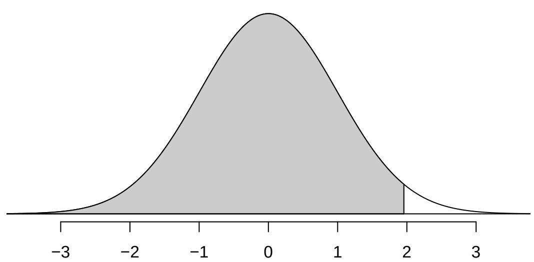
Example: Refer to the payday loan borrowers example from section 14.3.3. Estimate the true proportion of payday loan borrowers that would support the regulation.

Calculate the standard error of the sample proportion of payday loan borrowers that would support the regulation *when NOT assuming the null hypothesis is true*:

From the below, what multiplier ($z^\star$) should be used to create a 95% confidence interval for the parameter? Explain why 97.5% of the distribution is shaded (why are we using the 97.5th percentile to create a 95% confidence interval?)

For a 95% confidence interval:

```
openintro::normTail(m = 0, s = 1, L=1.96)
qnorm(0.975, lower.tail=TRUE)
#> [1] 1.959964
```



Calculate the margin of error for the 95% confidence interval from the payday loan borrowers data:

Calculate the 95% confidence interval from the payday loan borrowers data:

Confidence interval interpretation:

- How confident you are (e.g., 90%, 95%, 98%, 99%)
- Parameter of interest (including context: observational units, variable (if variable is categorical, define "success"))
- Calculated interval

4.2.1.1 Interpreting confidence level

What does it mean to be 95% confident in a created confidence interval?

- Our goal is to only take one sample from the population to create a confidence interval.
- Based on the 68-95-99.7 rule, we know that approximately ________% of sample ____________________ will fall within ______________ from the parameter.
- If we create 95% confidence intervals, ___________% of samples will create a 95% ____________________ interval that will contain the ___________________ of interest.
- 95% of samples accurately ____________________ the parameter of interest
  - When we create one confidence interval, we are 95% _______________________ that we have a "good" sample that created a confidence interval that contains the ________________ of interest.

4.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. What conditions must be met to use the Normal Distribution to approximate the sampling distribution of sampling proportions?

2. Should the conclusion include a population word like *true* or *long-run*? Explain your answer.

4.3 Activity 7: Handedness of Professional Male Boxers

4.3.1 Learning outcomes

- Describe and perform a theory-based hypothesis test for a single proportion.
- Check the appropriate conditions to use a theory-based hypothesis test.
- Calculate and interpret the standardized sample proportion.
- Interpret and evaluate a p-value for a theory-based hypothesis test for a single proportion.
- Use the standard normal distribution to find the p-value.

4.3.2 Terminology review

In this activity, we will introduce theory-based hypothesis tests for a single categorical variable. Some terms covered in this activity are:

- Parameter of interest
- Standardized statistic
- Normal distribution
- p-value

To review these concepts, see Chapter 11 & 14 in your textbook.

Activities from module 3 covered simulation-based methods for hypothesis tests involving a single categorical variable. This activity covers theory-based methods for testing a single categorical variable.

4.3.3 Handedness of male boxers

Left-handedness is a trait that is found in about 10% of the general population. Past studies have shown that left-handed men are over-represented among professional boxers (Richardson and Gilman 2019). Is there evidence that there is an over-prevalence of left-handed fighters? In this random sample of 500 professional male boxers, 81 were left-handed.

- Observational units:
- Variable:
  - Success:

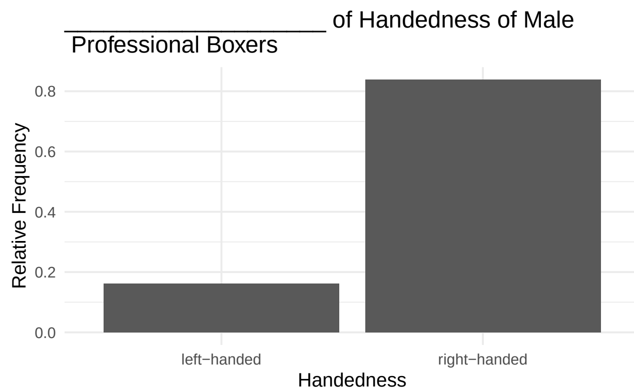
R Instructions

- Download the R file for today's activity from Canvas
- Upload the file to the R server
- Run lines 1–15 to load the needed packages and the data set and create a plot of the data

```
 # Read in data set
boxers <- read.csv("https://math.montana.edu/courses/s216/data/Male_boxers_sample.csv")
boxers %>% count(Stance)  # Count number in each Stance category
```

```
#>        Stance   n
#> 1  left-handed  81
#> 2 right-handed 419
```

```
boxers %>% # Data set piped into...
    ggplot(aes(x = Stance)) +    # This specifies the variable
    geom_bar(aes(y = after_stat(prop), group = 1)) +  # Tell it to make a bar plot with proportions
    labs(title = "____________________ of Handedness of Male \n Professional Boxers",
       # Give your plot a title
       x = "Handedness",    # Label the x axis
       y = "Relative Frequency")  # Label the y axis
```



1. What type of plot was created of these data?

Hypotheses and summary statistics

2. Write out the parameter of interest in words, in context of the study.

3. Write out the null hypothesis **in words**.

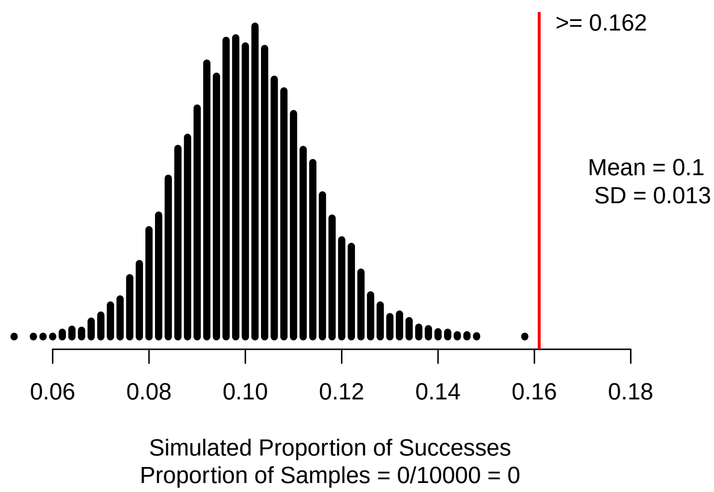
4. Write out the alternative hypothesis **in notation**.

5. Calculate the value of the summary statistic (sample proportion) for this study. Use proper notation.

Simulation methods

We could use simulation methods to analyze the research question. The following is the null distribution of 10,000 simulations.

```
one_proportion_test(probability_success = 0.1, # Null hypothesis value
          sample_size = 500, # Enter sample size
          number_repetitions = 10000, # Enter number of simulations
          as_extreme_as = 0.162, # Observed statistic
          direction = "greater", # Specify direction of alternative hypothesis
          summary_measure = "proportion") # Reporting proportion or number of successes?
```



In today's activity, we will use theory-based methods.

Theory-based methods

The sampling distribution of a single proportion — how that proportion varies from sample to sample — can be mathematically modeled using the normal distribution if certain conditions are met.

Conditions for the sampling distribution of $\hat{p}$ to follow an approximate normal distribution:

- **Independence**: The sample's observations are independent. (*Remember*: This also must be true to use simulation methods!)
- **Large enough sample size**: Success-failure condition: We *expect* to see at least 10 successes and 10 failures in the sample, $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$.

- Verify that the independence condition for the male professional boxer study is satisfied.

- Verify that the sample size for the male professional boxer study is large enough.

To calculate the standardized statistic we use the general formula

$$Z = \frac{\text{point estimate} - \text{null value}}{SE_0(\text{point estimate})}.$$

For a single categorical variable the standardized sample proportion is calculated using

$$Z = \frac{\hat{p} - \pi_0}{SE_0(\hat{p})},$$

where the standard error is calculated using the null value:

$$SE_0(\hat{p}) = \sqrt{\frac{\pi_0 \times (1 - \pi_0)}{n}}$$

.

For this study, the null standard error of the sample proportion is calculated using the null value, 0.1.

$$SE_0(\hat{p}) = \sqrt{\frac{0.1 \times (1 - 0.1)}{500}} = 0.013$$

.

Each sample proportion of professional male boxers that are left-handed is 0.013 from the true proportion of professional male boxers that are left-handed, on average.

Label the standard normal distribution shown below with the null value as the center value (below the value of zero). Label the tick marks to the right of the null value by adding 1 standard error to the null value to represent 1 standard error, 2 standard errors, and 3 standard errors from the null. Repeat this process to the left of the null value by subtracting 1 standard error for each tick mark.

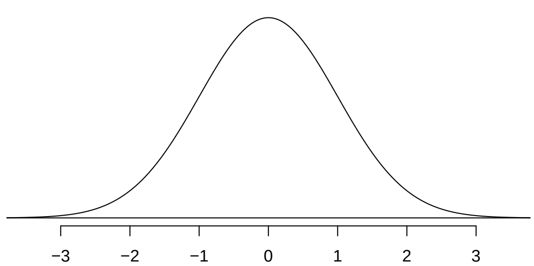


Figure 4.1: Standard Normal Curve

6. Using the null standard error of the sample proportion, calculate the standardized sample proportion (Z). Mark this value on the standard normal distribution shown.

The standardized statistic is used as a ruler to measure how far the sample statistic is from the null value. Essentially, we are converting the sample proportion into a measure of standard errors to compare to the standard normal distribution.

The standardized statistic measures the *number of standard errors the sample statistic is from the null value*.

Interpretation of the standardized sample proportion:

We will use the `pnorm()` function in `R` to find the p-value. In the code below, notice that we used `lower.tail = FALSE` to find the p-value. `R` will calculate the p-value *greater* than the value of the standardized statistic.

Notes:

- Use `lower.tail = TRUE` when doing a left-sided test.
- Use `lower.tail = FALSE` when doing a right-sided test.
- To find a two-sided p-value, use a left-sided test for negative Z or a right-sided test for positive Z, then multiply the value found by 2 to get the p-value.

```
pnorm(4.769, # Enter value of standardized statistic
      m=0, s=1, # Using the standard normal mean = 0, sd = 1
      lower.tail=FALSE) # Gives a p-value greater than the standardized statistic
```

```
#> [1] 9.257133e-07
```

7. Report the p-value obtained from the `R` output.

8. Write a conclusion based on the p-value.

9. To what group of observational units can the results be generalized to?

Two-sided test

Suppose that we want to show that the true proportion of professional male boxers **differs** from that in the general population.

10. Write out the alternative hypothesis in notation for this new research question.

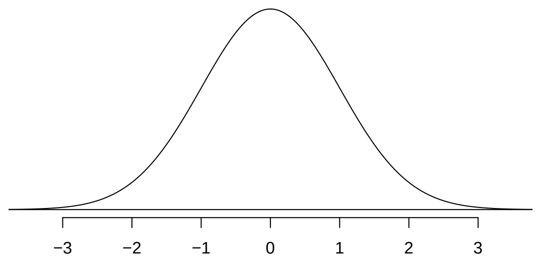


Figure 4.2: Standard Normal Curve

11. How would this impact the p-value? Would the p-value be larger or smaller?

4.3.4 Take-home messages

1. Both simulation and theory-based methods can be used to find a p-value for a hypothesis test. In order to use theory-based methods we need to check that both the independence and **the success-failure conditions** are met.
2. The standardized statistic measures how many standard errors the statistic is from the null value. The larger the standardized statistic the more evidence there is against the null hypothesis.
3. The p-value for a two-sided test is approximately two times the value for a one-sided test. A two-sided test provides less evidence against the null hypothesis.

4.3.5 Additional notes

Activities 7–9 cover the material in Module 4. For more practice with theory-based hypothesis testing of a single categorical variable use the Module 4 Review worksheet in the Unit 1 Review Materials (pgs. 117–121).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

4.4 Activity 8: Confidence intervals and what confidence means

4.4.1 Learning outcomes

- Explore what confidence means
- Interpret the confidence level

4.4.2 Terminology review

In this activity, we will explore what being 95% confidence means. Some terms covered in this activity are:

- Parameter of interest
- Two-sided vs. one-sided tests
- Confidence level

4.4.3 Handedness of male boxers continued

In today's activity, we will use the male boxer study to look at what confidence means.

Left-handedness is a trait that is found in about 10% of the general population. Past studies have shown that left-handed men are over-represented among professional boxers (Richardson and Gilman 2019). Is there evidence that there is an over-prevalence of left-handed fighters? In this random sample of 500 professional male boxers, 81 were left-handed.

```
# Read in data set
boxers <- read.csv("https://math.montana.edu/courses/s216/data/Male_boxers_sample.csv")
boxers %>% count(Stance)  # Count number in each Stance category
```

```
#>         Stance   n
#> 1  left-handed  81
#> 2 right-handed 419
```

What does *confidence* mean?

In the interpretation of a 95% confidence interval, we say that we are 95% confident that the parameter is within the confidence interval. Why are we able to make that claim? What does it mean to say "we are 95% confident"?

1. In the last activity we found very strong evidence that the true proportion of male professional boxers that are left-handed is greater than 0.1. As a class, determine a plausible value for the true proportion of professional male boxers that are left-handed. *Note: we are making assumptions about the population here. This is not based on our calculated data, but we will use this applet to better understand what happens when we take many, many samples from this believed population.*

2. Go to this website, http://www.rossmanchance.com/ISIapplets.html and choose 'Simulating Confidence Intervals'. In the input on the left-hand side of the screen enter the value from question 1 for π (the true value), 500 for n, and 100 for 'Number of intervals'. Click 'sample'.

- In the graph on the bottom right, click on a green dot. Write down the confidence interval for this sample given on the graph on the left. Does this confidence interval contain the true value chosen in question 1?

- Now click on a red dot. Write down the confidence interval for this sample. Does this confidence interval contain the true value chosen in question 1?

- How many intervals out of 100 contain π, the true value chosen in question 1? *Hint*: This is given to the left of the graph of green and red intervals.

3. Click on 'sample' nine more times. Write down the 'Running Total' for the proportion of intervals that contain π.

Interpretation of the level of confidence:

Notes on theory-based confidence intervals

To calculate a theory-based 95% confidence interval for π, we will first find the **standard error** of $\hat{p}$ by plugging in the value of $\hat{p}$ for π in $SD(\hat{p})$:

$$SE(\hat{p}) = \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}$$

Note that we do not include a "0" subscript, since we are not assuming a null hypothesis.

Recall from the last activity, the sample proportion is

$$\hat{p} = \frac{81}{500} = 0.162$$

.

Calculate the standard error of the sample proportion to find a 95% confidence interval.

We will calculate the margin of error and confidence interval later in this activity. **The margin of error (ME)** is the value of the z^* multiplier times the standard error of the statistic.

$$ME = z^* \times SE(\hat{p})$$

The z^* multiplier is the percentile of a standard normal distribution that corresponds to our confidence level. If our confidence level is 95%, we find the Z values that encompass the middle 95% of the standard normal distribution. If 95% of the standard normal distribution should be in the middle, that leaves 5% in the tails, or 2.5% in each tail.

The `qnorm()` function in R will tell us the z^* value for the desired percentile (in this case, $95\% + 2.5\% = 97.5\%$ percentile).

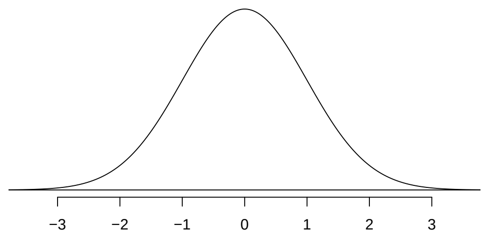


Figure 4.3: Standard Normal Curve

The following code will find the z^* value for a 95% confidence interval.

```
qnorm(c(0.025, 0.975), lower.tail = TRUE) # Multiplier for 95% confidence interval
#> [1] -1.959964  1.959964
```

Calculate the margin of error for the 95% confidence interval.

To find the confidence interval, we will add and subtract the **margin of error** to the point estimate:

$$\text{point estimate} \pm \text{margin of error}$$

$$\hat{p} \pm z^* \times SE(\hat{p})$$

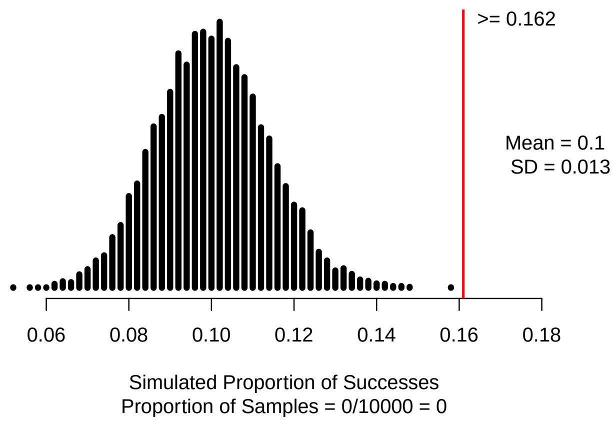
Calculate the 95% confidence interval for the parameter of interest.

4. Interpret the 95% confidence **interval** in the context of the problem.

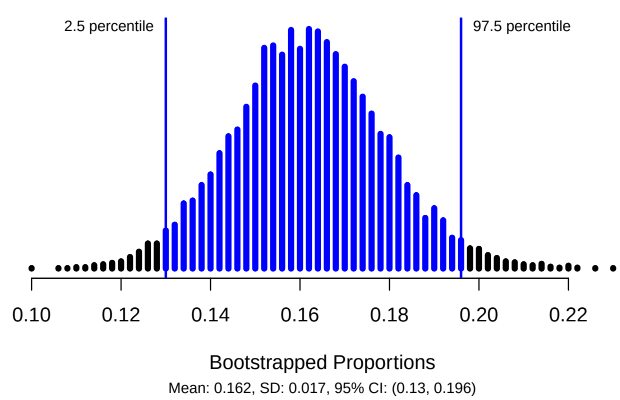
Simulation methods

We could also use simulation-based methods to analyze these data. Note the inputs into the R code to create the null and bootstrap distribution.

```
one_proportion_test(probability_success=0.1,
                    sample_size=500,
                    number_repetitions=10000,
                    as_extreme_as=0.162,
                    direction="greater",
                    summary_measure="proportion")
```




```
one_proportion_bootstrap_CI(sample_size = 500,
                            number_successes = 81,
                            number_repetitions = 10000,
                            confidence_level = 0.95)
```



5. Explain why the results for simulation methods and theory-based methods are similar.

4.4.4 Take-home messages

1. If repeat samples of the same size are selected from the population, approximately 95% of samples will create a 95% confidence interval that contains the parameter of interest.
2. The calculation of the confidence interval uses the standard error calculated using the sample proportion rather than the null value.
3. A confidence interval is built around the point estimate or observed calculated statistic from the sample. This means that the sample statistic is always the center of the confidence interval. A confidence interval includes a measure of sample to sample variability represented by the **margin of error**.

4.4.5 Additional notes

Activities 7–9 cover the material in Module 4. For more practice with theory-based confidence intervals for a single categorical variable use the Module 4 Review worksheet in the Unit 1 Review Materials (pgs. 117–121).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

4.5 Activity 9: Impacts on the P-value and Confidence Interval

4.5.1 Learning outcomes

- Explore impact of sample size, direction of the alternative hypothesis, and value of the sample statistic on the p-value.
- Explore impact of sample size and confidence level on the width of the confidence interval.

4.5.2 Terminology review

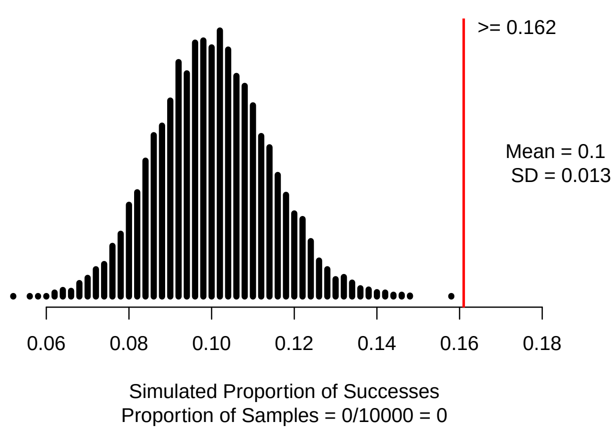
In today's activity, we will cover how different parameters may impact the p-value and confidence interval. Some terms covered in this activity are:

- Impacts of the sample statistic
- Impacts of sample size
- Effect of confidence level

Impacts on the p-value

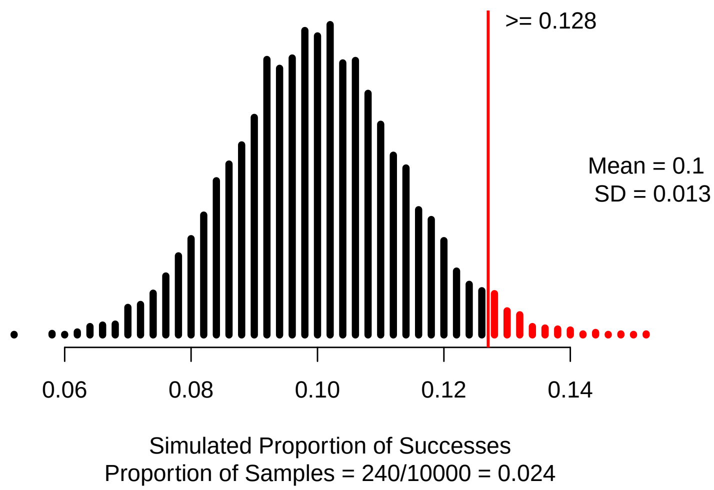
Let's see how each of the following with change the p-value. We will revisit the male professional boxer study. In this study, the researchers found that 81 out of 500 male professional boxers were left-handed. The following is the null distribution with 10,000 simulations assuming that the true proportion of male professional boxers that are left-handed is 0.1.

```
one_proportion_test(probability_success=0.1,
                    sample_size=500,
                    number_repetitions=10000,
                    as_extreme_as=0.162,
                    direction="greater",
                    summary_measure="proportion")
```



In another sample of 500 male professional boxers, only 64 selected the helper toy. The null distribution created below reflects this new sample.

```r
one_proportion_test(probability_success = 0.1, # Null hypothesis value
        sample_size = 500, # Enter sample size
        number_repetitions = 10000, # Enter number of simulations
        as_extreme_as = 0.128, # Observed statistic
        direction = "greater", # Specify direction of alternative hypothesis
        summary_measure = "proportion") # Reporting proportion or number of successes?
```



1. Note that this statistic ($\frac{64}{500} = 0.128$) is closer to the null value of 0.1 than the original statistic ($\frac{81}{500} = 0.1645$). Is the p-value for this new study larger or smaller than for the original statistic?

In activity 7, the following standardized proportion was calculated.

$$Z = \frac{0.164 - 0.1}{0.013} = 4.923$$

For this new study, the standard error for the sample proportion is the same.

$$SE_0(\hat{p}) = \sqrt{\frac{0.1(1-0.1)}{500}} = 0.013$$

2. Calculate the standardized statistic for the new study using a sample proportion of 0.128 ($\hat{p} = 0.128$).

3. Illustrate how you would find the p-value from the standardized statistic from this new study on the following standard normal distribution.

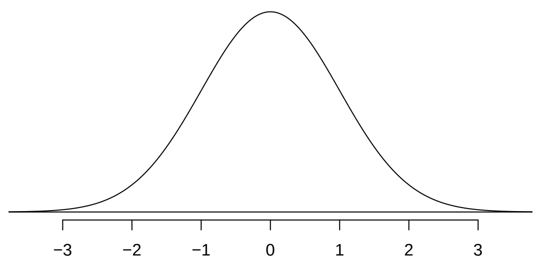


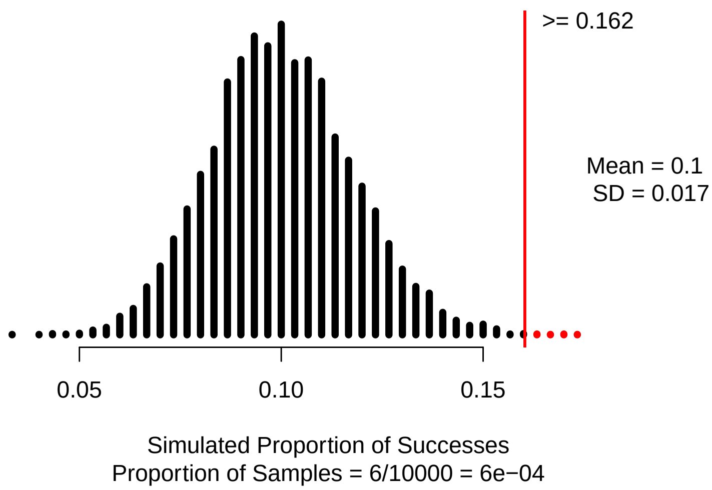
Figure 4.4: Standard Normal Distribution

Would this p-value be smaller or larger than the one found in Activity 7?

Change in sample size

In still another study, the researchers took a sample of 300 infants and found the same proportion ($\hat{p} = 0.162$) of male boxers that are left-handed. The null distribution created below reflects this new study.

```
one_proportion_test(probability_success = 0.1, # Null hypothesis value
          sample_size = 300, # Enter sample size
          number_repetitions = 10000, # Enter number of simulations
          as_extreme_as = 0.162, # Observed statistic
          direction = "greater", # Specify direction of alternative hypothesis
          summary_measure = "proportion") # Reporting proportion or number of successes?
```



4. Is the p-value found for the smaller sample size ($n = 300$) larger or smaller than the p-value found with the larger sample size ($n = 500$).

5. Summarize how each of the following affected the p-value:

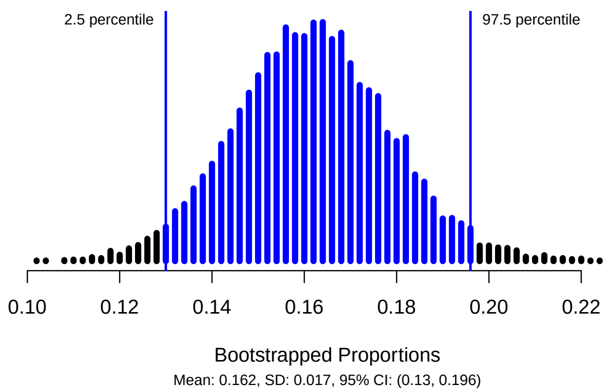
a) Using a larger sample size.

b) Using a sample statistic closer to the null value.

Effect of Confidence Level

To look at the impact on confidence interval, we will again look at the male professional boxer study. The following bootstrap distribution shows a 95% confidence interval for the parameter of interest.

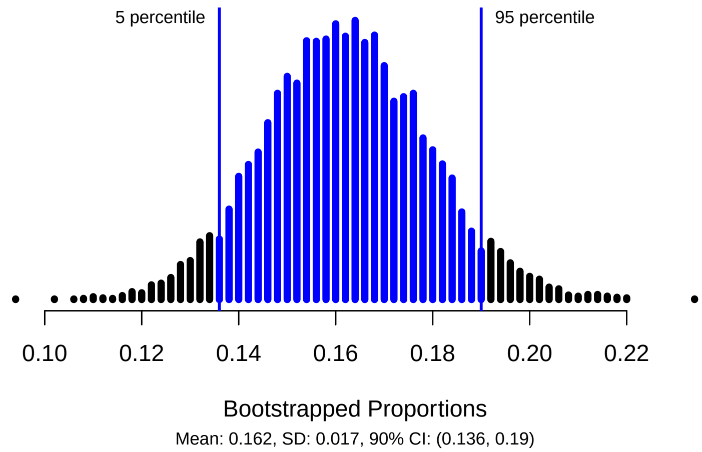
```
one_proportion_bootstrap_CI(sample_size = 500,
                            number_successes = 81,
                            number_repetitions = 10000,
                            confidence_level = 0.95)
```



6. Suppose instead of finding a 95% confidence interval, we found a 90% confidence interval. Would you expect the 90% confidence interval to be narrower or wider? Explain your answer.

7. The following R code produced the bootstrap distribution with 10000 simulations that follows. Circle the value that changed in the code.

```
one_proportion_bootstrap_CI(sample_size = 500, # Sample size
                    number_successes = 81, # Observed number of successes
                    number_repetitions = 10000, # Number of bootstrap samples to use
                    confidence_level = 0.90) # Confidence level as a decimal
```



8. Report both the 95% confidence interval and the 90% confidence interval (question 7). Is the 90% confidence interval narrower or wider than the 95% confidence interval?

Effect of sample size on the width of the confidence interval

How would an decrease in sample size impact the width of the confidence interval? We will again use a sample size of 300 male boxers that shows the same proportion ($\hat{p} = 0.162$) of male boxers that are left-handed.

The standard error of the sample proportion for this study with the smaller sample size is:

$$SE(\hat{p}) = \sqrt{\frac{0.162 \times (1 - 0.162)}{300}} = 0.0213$$

9. Is the standard error of the sample proportion for this study smaller or larger than the value calculated earlier?

Recall that the z^* multiplier is 1.96 for a 95% confidence interval.

10. Calculate the 95% confidence interval for this study with the smaller sample size.

The width of the confidence interval is found by calculating the difference between the upper value and the lower value.

$$\text{width of CI} = \text{upper CI value - lower CI value}$$

The center of the confidence interval is the sample statistic and can also be found using the midpoint of the interval.

$$\text{midpoint} = \frac{\text{lower CI value} + \text{upper CI value}}{2}$$

11. Compare the interval found in question 10 to the interval calculated in Activity 8 (prior to question 4).

- Calculate the center of the interval with the smaller sample size.
- Calculate the center of the interval from Activity 8.
- Did the center of the interval change?
- Calculate the width of the interval with the smaller sample size.
- Calculate the width of the interval from Activity 8.
- Which interval is wider?

The margin of error represents half the width of the confidence interval since we add and subtract the margin of error to the value of the sample statistic.

$$\text{width of CI} = 2 \times \text{ME}$$

12. Using the width of the interval with the smaller sample size calculated in question 11, calculate the margin of error.

13. What impact does decreasing the sample size have on the width of the confidence interval?

4.5.3 Take-home messages

1. The larger the sample size, the smaller the sample to sample variability. This will result in a larger standardized statistic and more evidence against the null hypothesis.
2. The farther the statistic is from the null value, the larger the standardized statistic. This will result in a smaller p-value and more evidence against the null hypothesis.
3. Decreasing the sample size, increases the sample to sample variability, the standard error resulting in a wider confidence interval.

4.5.4 Additional notes

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

4.6 Module 3 and 4 Lab: Mixed Breed Dogs in the U.S.

4.6.1 Learning outcomes

- Determine whether simulation or theory-based methods of inference can be used.
- Analyze and interpret a study involving a single categorical variable.

4.6.2 Mixed Breed Dogs in the U.S.

> The American Veterinary Medical Association estimated in 2010 that approximately 49% of dog owners in the U.S. own dogs that are classified as "mixed breed." As part of a larger 2022 international study (Banton 2022) about overall dog health, survey participants were asked, among other things, to report whether their dog was purebred or a mixed breed. Seven hundred and fifty (750) dog owners from the U.S. were recruited to complete an online survey via an email indicating they had been randomly selected by Qualtrics (an "experience management" company that specializes in surveys). Three hundred sixty-four (364) out of 675 respondents from the U.S. reported they owned a mixed breed dog. In the last decade, has the proportion of dog owners in the U.S. that own a mixed breed dog has changed from the value reported in 2010?

- Observational units:
- Variable:

```
* Success:
```

R Instructions

- Download the R script file and the data file (US_dogs.csv) from Canvas
- Upload both files to Canvas and open the R script file
- Enter the name of the dataset for datasetname.csv in line 6
- Highlight and run lines 1 - 6 to load the data set

```r
dogs <- read.csv("datasetname.csv")
```

Use statistical analysis methods to draw inferences from the data

1. Write out the parameter of interest in words, in context of the study.

2. Write out the null and alternative hypotheses in notation.

   H_0 :

   H_A :

3. Will theory-based methods give the same results as simulation based methods? Explain your answer.

Exploratory Data Analysis

4. What is the value of the point estimate? Report the value with the appropriate notation.

5. Create a plot of the data using the R code. Make sure to include an appropriate title with type of plot, observational units, and variable.

```
dogs %>% # Data set piped into...
    ggplot(aes(x = variable)) +   # This specifies the variable
    geom_bar(aes(y = after_stat(prop), group = 1)) +  # Tell it to make a bar plot with proportions
    labs(title = "Don't forget to title your plot",
       # Give your plot a title
       x = "Don't forget an axis label!",   # Label the x axis
       y = "Relative Frequency")  # Label the y axis
```

Null Distribution

To use the computer simulation, we will need to enter the

- assumed "probability of success" (π_0),
- "sample size" (the number of observational units or cases in the sample),
- "number of repetitions" (the number of samples to be generated),
- "as extreme as" (the observed statistic), and
- the "direction" (matches the direction of the alternative hypothesis).

We will use the `one_proportion_test()` function in `R` (in the `catstats` package) to simulate the null distribution of sample proportions and compute a p-value.

- Using the provided `R` script file, fill in the values/words for each `xx` in the one proportion test to create a null distribution with 10000 simulations.
- Then highlight and run lines 21–26.

```
one_proportion_test(probability_success = xx, # Null hypothesis value
          sample_size = xx, # Enter sample size
          number_repetitions = 10000, # Enter number of simulations
          as_extreme_as = xx, # Observed statistic
          direction = "xx", # Specify direction of alternative hypothesis
          summary_measure = "proportion") # Reporting proportion or number of successes?
```

6. Report the p-value from the study.

Bootstrap distribution

We will use the `one_proportion_bootstrap_CI()` function in R (in the `catstats` package) to simulate the bootstrap distribution of sample proportions and calculate a confidence interval. Using the provided R script file, fill in the values/words for each `xx` in the one proportion bootstrap confidence interval (CI) code to create a bootstrap distribution with 10000 simulations. Then highlight and run lines 31–34 to create a 90% confidence interval.

```
one_proportion_bootstrap_CI(sample_size = xx, # Sample size
                    number_successes = xx, # Observed number of successes
                    number_repetitions = 10000, # Number of bootstrap samples to use
                    confidence_level = xx) # Confidence level as a decimal
```

7. Report the 90% confidence interval.

Summarize the results of the study

8. Write a paragraph summarizing the results of the study. Be sure to describe:

- Introduction statement
  - Research question
  - Observational units
  - Variable (type)
  - Sampling method
- Summary statistic and interpretation
  - Summary measure (in context)
  - Value of the statistic
- P-value and interpretation
  - Statement about probability or proportion of samples
  - Statistic (summary measure and value)
  - Direction of the alternative
  - Null hypothesis (in context)
- Confidence interval and interpretation
  - How confident you are (e.g., 90%, 95%, 98%, 99%)
  - Parameter of interest
  - Calculated interval
  - Order of subtraction when comparing two groups
- Conclusion (written to answer the research question)
  - Amount of evidence
  - Parameter of interest
  - Direction of the alternative hypothesis
- Scope of inference

- To what group of observational units do the results apply (target population or observational units similar to the sample)?

Upload a copy of your group's paragraph to Gradescope.

Paragraph:

MODULE 5

Unit 1 Review

The following module contains both a list of key topics covered in Unit 1 as well as Module Review Worksheets that will be covered in Weekly Review Sessions.

5.0.1 Key Topics

Review the key topics for Unit 1 prior to the first exams. All of these topics will be covered in Modules 1–4.

5.0.2 Module Review

The following worksheets review each of the modules. These worksheets will be completed during Melinda's Study Sessions each week. Solutions will be posted on Canvas in the Unit 1 Review folder after the study sessions.

5.1 Key Topics Exam 1

Descriptive statistics and study design

1. Identify the observational units.
2. Identify the types of variables (categorical or quantitative).
3. Identify the explanatory variable (if present) and the response variable (roles of variables).
4. Identify the appropriate type of graph and summary measure.
5. Identify if a given value is a statistic or a parameter. Identify the appropriate notation.
6. Identify the study design (observational study or randomized experiment).
7. Identify the sampling method and potential types of sampling bias (non-response, response, selection).
8. Identify and interpret the summary statistic
9. Identify the target population
10. Identify the types of sampling bias (response, non-response, selection, none)
11. Identify the type(s) of graph(s) that could be used to plot the given variable(s).

Hypothesis testing

12. Write the parameter of interest in context of the problem.
13. State the null and alternative hypotheses in both words and notation
14. Verify the validity condition is met to use simulation-based methods to find a p-value.
15. Verify the validity conditions are met to use theory-based methods to find a p-value from the theoretical distribution.
16. In a simulation-based hypothesis test, describe how to create one dot on a dotplot of the null distribution using coins, cards, or spinners.
17. Explain where the null distribution is centered and why.
18. Describe and illustrate how R calculates the p-value for a simulation-based test.
19. Describe and illustrate how R calculates the p-value for a theory-based test.
20. Type of theoretical distribution (standard normal distribution or t-distribution with appropriate degrees of freedom) used to model the standardized statistic in a theory-based hypothesis test.
21. Calculate and interpret the standard error of the statistic under the null using the correct formula on the Golden ticket.
22. Calculate and interpret the appropriate standardized statistic using the correct formula on the Golden ticket.
23. Interpret the p-value in context of the study: it is the probability of ______, assuming ______.
24. Evaluate the p-value for strength of evidence against the null: how much evidence does the p-value provide against the null?
25. Write a conclusion about the research question based on the p-value.
26. Describe which features of the study impact the p-value and how.

5.1.1 Confidence intervals

27. Describe how to simulate one bootstrapped sample using cards.
28. Explain where the bootstrap distribution is centered and why.
29. Find an appropriate percentile confidence interval using a bootstrap distribution from R output.
30. Verify the validity condition is met to use simulation-based methods to find the confidence interval.
31. Verify the validity conditions are met to use theory-based methods to calculate a confidence interval.
32. Describe and illustrate how the bootstrap distribution is used to find the confidence interval for a given confidence level.
33. Describe and illustrate how the standard normal distribution or t-distribution is used to find the multiplier for a given confidence level.
34. Calculate and interpret the standard error of the statistic (not assuming the null hypothesis) using the correct formula on the Golden ticket
35. Calculate the appropriate margin of error and confidence interval using theory-based methods.
36. Interpret the confidence interval in context of the study.
37. Based on the interval, what decision can you make about the null hypothesis? Does the confidence interval agree with the results of the hypothesis test? Justify your answer.
38. Interpret the confidence level in context of the study. What does "confidence" mean?
39. Describe which features of the study have an effect on the width of the confidence interval and how.

5.1.2 Probability

40. Calculate probabilities from a given table and give appropriate probability notation for both conditional and unconditional probabilities.
41. Create a two-way table using given probabilities.
42. Interpret a probability value in context of the problem.

5.2 Module 1 Review - Sampling Methods

1. Suppose that the proportion of all American adults that fit the medical definition of being obese is 0.23. A large medical clinic would like to determine if the proportion of their patients that are obese is higher than that of all American adults. The clinic takes a simple random sample of 30 of their patients and finds that 9 patients in the sample are obese.

a. What is the target population?

b. What are the observational units?

c. What variable is being studied?

d. Is the variable identified in part (c) categorical or quantitative?

2. Martha works in Macy's advertising department. She is interested in the shopping experience of all Macy's shoppers in the U.S. Every Saturday morning for a month she stands outside of the Bozeman Macy's asking people about their experience. One of the questions she uses is: "As a huge fan of Macy's, I believe Macy's has the best choices of clothing in Bozeman. Don't you agree?" Every person that was asked, responded.

a. Identify the target population.

b. Identify the sample.

c. Which of the three types of sampling bias (selection, non-response, response) may be present? Explain your choice(s).

3. This study aims to explore whether Swiss university students feeling academic study pressure (whether the student had experienced academic failure) tend to use psychotropic drugs (whether the student had used psychotropic drugs during the student's time at university) as a coping mechanism. An invitation email was sent to all bachelor's and master's students at the University of Lausanne, totaling 15,400 individuals, with a link to access the online questionnaire containing 49 questions and 107 items. No reminder was sent out, and no incentive was given to complete the questionnaire. A total of 1,690 students initially participated in the study, but 424 questionnaires were too incomplete to be used for analysis and were excluded. Additionally, 67 questionnaires were removed because of significant missing sociodemographic information, resulting in 1,199 completed responses included in the final analysis. Is there an association between study pressure and use of psychotropic drugs among Swiss University students?

a. Identify the target population.

b. Identify the sample.

c. Which of the three types of sampling bias (selection, non-response, response) may be present? Explain your choice(s).

d. Identify the type and roles of each variable in the study.

4. Researchers decided to investigate whether a cat's coat color is associated with aggressive cat behavior by creating a 20-minute survey. The survey was distributed by posting it to social media and through cat-related listservs (e.g., For the Love of Cats), inviting individuals to take the survey. A total of 1,365 surveys were completed by participants. The frequency of each of the following aggressive behavior categories was assessed: hiss, stalk/chase, bite, slap/scratch. Frequency of behaviors toward people were recorded on a 6-point scale: 0 = never, 1 = less than once every 6 months, 2 = once every 6 months, 3 = once per month, 4 = once per week, 5 = one or more times per day. Because there were four aggressive behavior categories, each with a frequency of 0 to 5 possible, each cat could score between 0 to 20 for human aggression. Is there an association between coat color and aggressive behavior among cats?

a. Identify the target population.

b. Identify the sample.

c. Which of the three types of sampling bias (selection, non-response, response) may be present? Explain your choice(s).

d. Identify the type and roles of each variable in the study.

5.3 Module 2 Review - Probability

1. Spelling errors in a text can either be non-word errors (teh instead of the) or word errors (lose instead of loose). It was found that non-word errors make up about 25% of all errors. A human proofreader will catch 92% of non-word errors and 75% of word errors.

   Let N represent non-word errors and C represent that a human proofreader will catch the error.

a. Identify the following values with appropriate probability notation.

   0.25

   0.92

   0.75

b. Fill in the table below to represent the situation:

	N	N^C	Total
C			
C^C			
Total			100000

c. Using your table calculate the probability that a randomly selected error caught by a human proofreader is a non-word error. Use appropriate probability notation.

d. Find the probability a selected error is a non-word error and was not caught by a human proofreader. Use appropriate probability notation.

e. Find the value of $P(N|C)$. What does this probability mean?

2. A private college report contains these statistics:

- 70% of incoming freshmen attended public schools
- 75% of public-school students who enroll as freshmen eventually graduate
- 90% of other freshmen eventually graduate

Let A represent the event that a freshman attended public school and B the event that a freshman eventually graduates.

a. Identify the following values with appropriate probability notation.

0.70

0.75

0.90

b. Fill in the table below to represent the situation:

	A	A^C	Total
B			
B^C			
Total			100000

c. Calculate the probability a selected freshman attended public school given they did not graduate. Use appropriate probability notation.

d. Calculate the probability a selected freshman does not graduate. Use appropriate probability notation.

e. Of the population of freshman that attended public school, what is the probability they do not graduate. Use appropriate probability notation.

f. Find the value of $P(A \text{ and } B^C)$. Write this probability in context of the problem.

5.4 Module 3 Review - Simulation Methods for a Single Proportion

A recent study examined hearing loss data for 1753 U.S. teenagers. In this sample, 328 were found to have some level of hearing loss. News of this study spread quickly, with many news articles blaming the prevalence of hearing loss on the higher use of ear buds by teens. At MSNBC.com (8/17/2010), Carla Johnson summarized the study with the headline: "1 in 5 U.S. teens has hearing loss, study says." Is this an appropriate or a misleading headline?

1. Write the parameter of interest in context of the study.

2. Write the null hypothesis in words and notation in context of the problem.

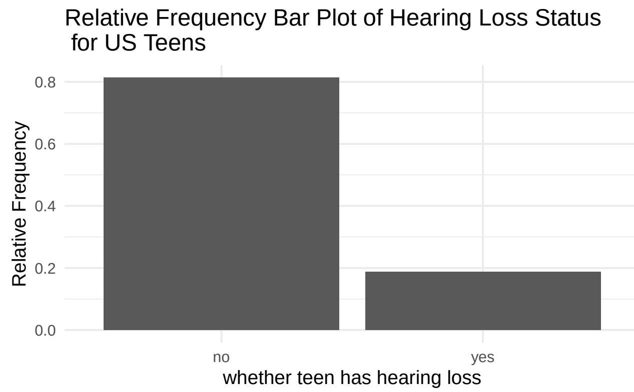
3. Based on the research questions, choose the direction for the alternative hypothesis.

4. Write the alternative hypothesis in words and notation in context of the problem.

5. Calculate the summary statistic. Use proper notation.

The following is a plot of the data.

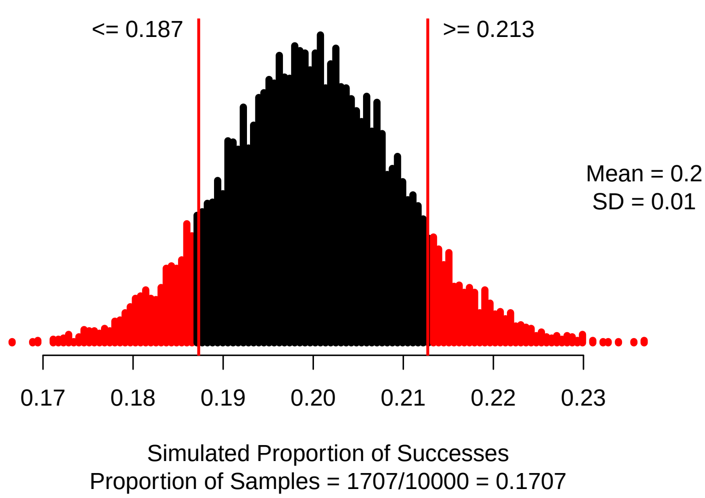
```
hearing %>% # Data set piped into...
    ggplot(aes(x = hearing_loss)) +   # This specifies the variable
    geom_bar(aes(y = after_stat(prop), group = 1)) +  # Tell it to make a bar plot with proportions
    labs(title = "Relative Frequency Bar Plot of Hearing Loss Status \n for US Teens",
       # Give your plot a title
       x = "whether teen has hearing loss",   # Label the x axis
       y = "Relative Frequency")  # Label the y axis
```



6. What values should be entered for each of the following into the one proportion test to create 10000 simulations?

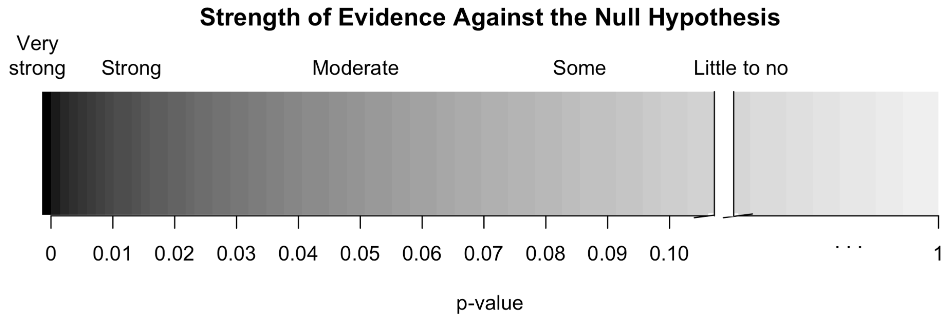
- Probability of success:
- Sample size:
- Number of repetitions:
- As extreme as:
- Direction (“greater”, “less”, or “two-sided”):

```
one_proportion_test(probability_success = 0.2, #Null hypothesis value
                    sample_size = 1753, #Enter sample size
                    number_repetitions = 10000, #Enter number of simulations
                    as_extreme_as = 0.187, #observed statistic
                    direction = "two-sided", #specify direction of alternative hypothesis
                    summary_measure = "proportion") #Reporting proportion or number of successes?
```



7. Interpret the p-value in context of the problem.

8. How much evidence does the data provide against the null hypothesis?



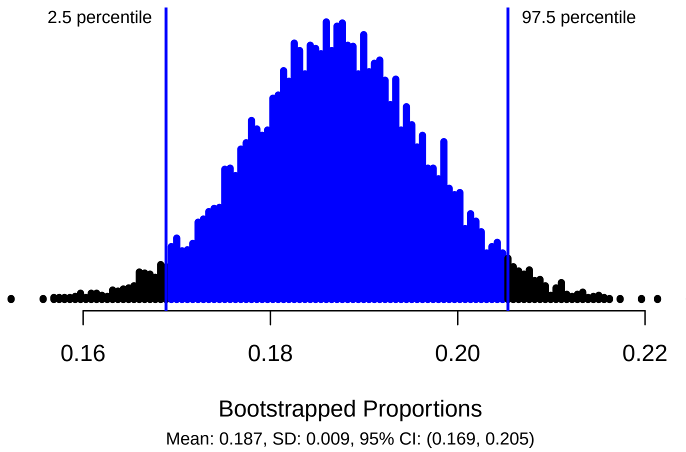
9. Write a conclusion to the study in context of the problem.

10. Would a 95% confidence interval contain the null value of 0.2? Explain.

11. What values should be entered for each of the following into the simulation to create the bootstrap distribution of sample proportions to find a 95% confidence interval?

- Sample size:
- Number of successes:
- Number of repetitions:
- Confidence level (as a decimal):

```
one_proportion_bootstrap_CI(sample_size = 1753, # Sample size
                    number_successes = 328, # Observed number of successes
                    number_repetitions = 10000, # Number of bootstrap samples to use
                    confidence_level = 0.95) # Confidence level as a decimal
```



12. Explain how to use cards to create one bootstrap sample.

13. Report the 95% confidence interval in interval notation.

14. Interpret the 95% confidence interval in context of the problem.

5.5 Module 4 Review - Theory-based Methods for a Single Proportion

Statistician Jessica Utts has conducted an extensive analysis of Ganzfeld studies that have investigated psychic functioning. Ganzfeld studies involve a "sender" and a "receiver." Two people are placed in separate rooms. The sender looks at a "target" image on a television screen and attempts to transmit information about the target to the receiver. The receiver is then shown four possible choices or targets, one of which is the correct target and the other three are "decoys." The receiver must choose the one he or she thinks best matches the description transmitted by the sender. If the correct target is chosen by the receiver, the session is a "hit." Otherwise, it is a miss. Utts reported that her analysis considered a total of 2,124 sessions and found a total of 709 "hits" (Utts, 2010). Is there evidence of psychic ability?

1. Write the parameter of interest in context of the study.

2. Calculate the point estimate. Use proper notation.

3. Write the null hypothesis in words.

4. Write the alternative hypothesis in notation.

A single proportion can be mathematically modeled using the normal distribution if certain conditions are met.

Conditions for the sample distribution of $\hat{p}$.

- Independence: The sample's observations are independent, e.g., are from a simple random sample
- Large enough sample size:
  - Success-Failure Condition: There are at least 10 successes and 10 failures in the sample

$$n \times \hat{p} \geq 10$$

and

$$n \times (1 - \hat{p}) \geq 10$$

5. Are the conditions met to model the data with the Normal distribution?

Standardized sample proportion.

The standardized statistic for theory-based methods for one proportion is:

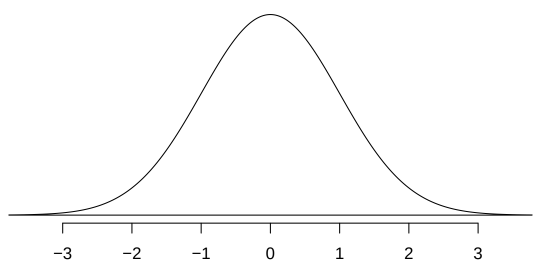
$$Z = \frac{\hat{p} - \pi_0}{SE_0(\hat{p})}$$

Where

$$SE_0(\hat{p}) = \sqrt{\frac{\pi_0 \times (1-\pi_0)}{n}}$$

6. Calculate the null standard error of the sample proportion

7. Calculate the standardized statistic for the sample proportion.



8. Interpret the standardized statistic in context of the problem.

We will use the `pnorm()` function in `R` to find the p-value. The value of the standardized statistic calculated in question 8 is entered into the `R` code. We used `lower.tail = FALSE` to find the p-value so that `R` will calculate the p-value *greater* than the value of the standardized statistic.

Notes:

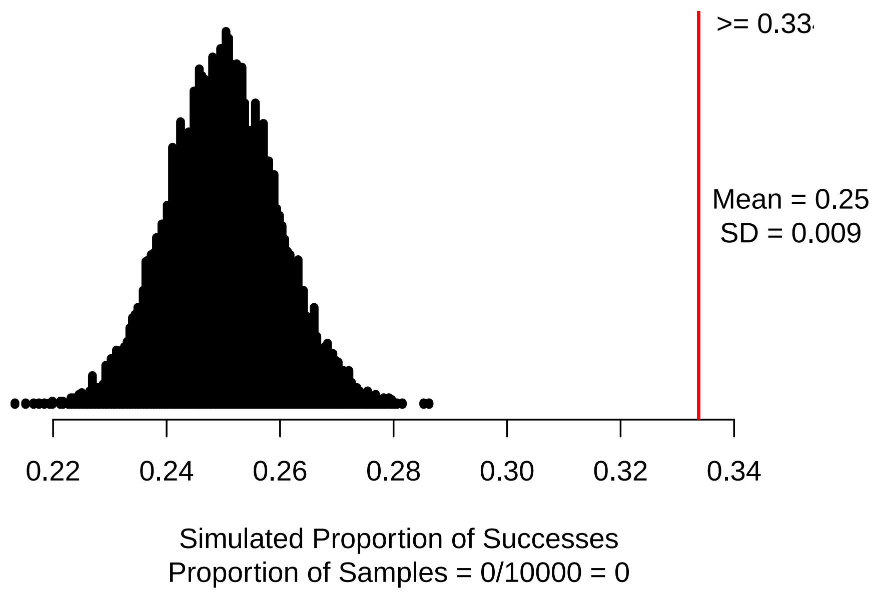
- Use `lower.tail = TRUE` when doing a left-sided test.
- Use `lower.tail = FALSE` when doing a right-sided test.
- To find a two-sided p-value, use a left-sided test for negative Z or a right-sided test for positive Z, then multiply the value found by 2 to get the p-value.

```
pnorm(9.333, # Enter value of standardized statistic
      m=0, s=1, # Using the standard normal mean = 0, sd = 1
      lower.tail=FALSE) # Gives a p-value greater than the standardized statistic
#> [1] 5.145792e-21
```

9. Report the value of the p-value.

Simulation Method:

```
one_proportion_test(probability_success = 0.25, #Null hypothesis value
                    sample_size = 2124, #Enter sample size
                    number_repetitions = 10000, #Enter number of simulations
                    as_extreme_as = 0.334, #observed statistic
                    direction = "greater", #specify direction of alternative hypothesis
                    summary_measure = "proportion") #Reporting proportion or number of successes?
```



10. Interpret the p-value in context of the study.

Next we will use theory-based methods to estimate the parameter of interest.

To calculate a theory-based 95% confidence interval for π, we will first find the **standard error** of $\hat{p}$ by plugging in the value of $\hat{p}$ for π in $SD(\hat{p})$:

$$SE(\hat{p}) = \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}.$$

Note that we do not include a "0" subscript, since we are not assuming a null hypothesis.

11. Calculate the standard error of the sample proportion to find a 95% confidence interval.

To find the confidence interval, we will add and subtract the **margin of error** to the point estimate:

$$\text{point estimate} \pm \text{margin of error}$$
$$\hat{p} \pm z^* SE(\hat{p})$$

The z^* multiplier is the percentile of a standard normal distribution that corresponds to our confidence level. If our confidence level is 95%, we find the Z values that encompass the middle 95% of the standard normal distribution. If 95% of the standard normal distribution should be in the middle, that leaves 5% in the tails, or 2.5% in each tail. The `qnorm()` function in `R` will tell us the z^* value for the desired percentile (in this case, 95% + 2.5% = 97.5% percentile).

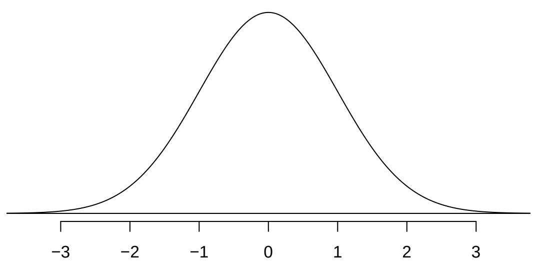


Figure 5.1: A standard normal curve.

```
qnorm(0.975) # Multiplier for 95% confidence interval
```

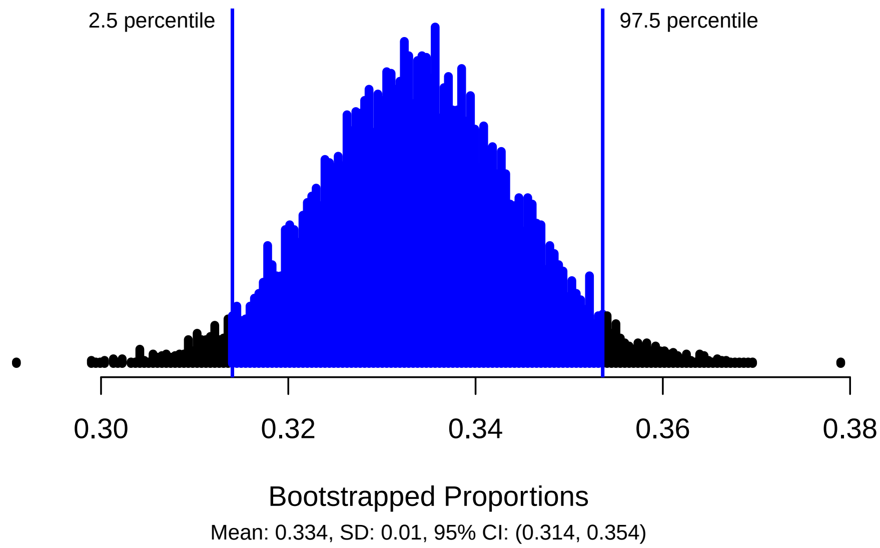
```
#> [1] 1.959964
```

12. Calculate the margin of error for a 95% confidence interval for the true proportion of sessions that will result in a hit.

13. Calculate the 95% confidence interval for the true proportion of sessions that will result in a hit.

Simulation Methods:

```
one_proportion_bootstrap_CI(sample_size = 2124, # Sample size
                    number_successes = 709, # Observed number of successes
                    number_repetitions = 10000, # Number of bootstrap samples to use
                    confidence_level = 0.95) # Confidence level as a decimal
```



14. Interpret the 95% confidence interval in context of the problem.

15. Write a conclusion based on the p-value and the 95% confidence interval.

5.6 Group Exam 1 Review

Use the provided data set from the Islands (Bulmer, n.d.) (Exam1ReviewData.csv) and the appropriate Exam 1 Review R script file to answer the following questions. Each adult (>21) islander was selected at random from all adult islanders. Note that some islanders choose not to participate in the study. These islanders that did not consent to be in the study are removed from the dataset before analysis. Variables and their descriptions are listed below. Here is some more information about some of the variables collected. Music type (classical or heavy metal) was randomly assigned to the Islanders. Time to complete the puzzle cube was measured after listening to music for each Islander. Heart rate and blood glucose levels were both measured before and then after drinking a caffeinated beverage.

Variable	**Description**
`Island`	Name of Island that the Islander resides on
`City`	Name of City in which the Islander resides
`Population`	Population of the City
`Name`	Name of Islander
`Consent`	Whether the Islander consented to be in the study (`Declined`, `Consented`)
`Gender`	Gender of Islander (`M` = male, `F` = Female)
`Age`	Age of Islander
`Married`	Marital status of Islander (`yes`, `no`)
`Smoking_Status`	Whether the Islander is a current smoker (`nonsmoker`, `smoker`)
`Children`	Whether the Islander has children (`yes`, `no`)
`weight_kg`	Weight measured in kg
`height_cm`	Height measured in cm
`respiratory_rate`	Breaths per minute
`Type_of_Music`	Music type Islander was randomly assigned to listen to (`Classical`, `Heavy Metal`)
`After_PuzzleCube`	Time to complete puzzle cube (minutes) after listening to assigned music
`Education_Level`	Highest level of education completed (`highschool`, `university`)
`Balance_Test`	Time balanced measured in seconds with eyes closed
`Blood_Glucose_before`	Level of blood glucose (mg/dL) before consuming assigned drink
`Heart_Rate_before`	Heart rate (bpm) before consuming assigned drink
`Blood_Glucose_after`	Level of blood glucose (mg/dL) after consuming assigned drink
`Heart_Rate_after`	Heart rate (bpm) after consuming assigned drink
`Diff_Heart_Rate`	Difference in heart rate (bpm) for Before - After consuming assigned drink
`Diff_Blood_Glucose`	Difference in blood glucose (mg/dL) for Before - After consuming assigned drink

1. What are the observational units?

2. In the table above, indicate which variables are categorical (C) and which variables are quantitative (Q).

3. What type of sampling bias may be present in this study? Explain.

4. Use the appropriate Exam 1 Review R script file to find the summary statistic and graphical display of the data to assess the following research question, "Is there evidence that the proportion of adult Islanders that smoke differs from the reported value of 11%?"

a. What is the name of the variable to be assessed in this research question?

What type of variable (categorical or quantitative) is the variable you identified?

b. Parameter of Interest:

c. Null Hypothesis:

Notation:

Words:

d. Alternative Hypothesis:

Notation:

Words:

e. Use the R script file to get the counts for each level of the variable. Fill in the following table with the variable name, levels of the variable, and counts using the values from the R output.

	Count
Success	
Failure	
Total	

f. Calculate the value of the summary statistic to answer the research question. Give appropriate notation.

g. What type of graph(s) would be appropriate for this research question?

h. Using the provided R file create a graph of the data. Sketch the graph below:

i. Assess if the following conditions are met:

Independence (needed for both simulation and theory-based methods):

Success-Failure (must be met to use theory-based methods):

j. Use the provided R script file to find the simulation p-value to assess the research question. Report the p-value.

k. Interpret the p-value in the context of the problem.

l. Write a conclusion to the research question based on the p-value.

m. Using a significance level of $\alpha = 0.05$, what statistical decision will you make about the null hypothesis?

n. Use the provided R script file to find a 95% confidence interval.

o. Interpret the 95% confidence interval in context of the problem.

p. Regardless to your answer in part i, calculate the standardized statistic.

q. Interpret the value of the standardized statistic in context of the problem.

r. Use the provided R script file to find the theory-based p-value.

s. Use the provided R script file to find the appropriate z* multiplier and calculate the theory-based confidence interval.

t. Does the theory-based p-value and CI match those found using simulation methods? Explain why or why not.

u. To what group of observational units do the results apply?

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